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Dynamic Programming Applications in Water Resources

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The central intention of this survey is to review dynamic programming models for water resource problems and to examine computational techniques which have been used to obtain solutions to these problems. Problem areas surveyed here include aqueduct design, irrigation system control, project development, water quality maintenance, and reservoir operations analysis. Computational considerations impose severe limitation on the scale of dynamic programming problems which can be solved. Inventive numerical techniques for implementing dynamic programming have been applied to water resource problems. Discrete dynamic programming, differential dynamic programming, state incremental dynamic programming, and Howard's policy iteration method are among the techniques reviewed. Attempts have been made to delineate the successful applications, and speculative ideas are offered toward attacking problems which have not been solved satisfactorily.

CONTENTS

Introduction and orientation 673
Notation, perspectives, and preliminaries 673
Hydrologic models 675
  Water quality maintenance in rivers and sewers 675
  Water resource project development 676
  Irrigation management 678
  Other applications 680
Reservoir operations 682
  Deterministic case 682
  Stochastic case 685
Computational aspects 688
  Deterministic case 688
  Stochastic case 691
Abbreviated list of symbols 694

1. INTRODUCTION AND ORIENTATION

This survey is divided into six sections. It is long and diverse; few readers will be interested in all its topics. With the exception of the fundamental section 2, the survey is written to assist those who would skip and scan. Hopefully, the subsequent parts can be read in any order. Toward helping the reader devise a plan, a brief synopsis of the contents and the intentions of this survey is now presented.

In section 2, concepts and methods of dynamic programming are set forth. In particular, precise definitions for notions such as 'state,' 'policy,' and 'loss function' will serve to simplify and unify subsequent descriptions of dynamic programming applications to water resource problems. Also, reading suggestions for background and introduction to advanced methodology for dynamic programming will be offered.

Section 3 is devoted to a survey of dynamic programming models in the areas of water quality management, water resource project development, irrigation management, and aqueduct design. Through examination of the dynamic programming structure one can note similarities and arrive at conclusions concerning the relative computational requirements of different models.

The literature of reservoir operations analysis is as vast as that of the combined topics in section 3. Thus all of section 4 is devoted to this subject. It would seem that an excellent opportunity exists for development of 'scientific' operation curves; but for reasons pointed out in sections 4 and 5, this opportunity has not been fully realized.

In section 5, attention shifts from modelling to computational aspects of dynamic programming for water resource problems. For reasons quite familiar to those involved in numerical solution of partial differential equations, the most obvious computational device of simply discretizing the state space leads to an exponential growth in computational effort with increase in state dimension. The practical consequence is that for dynamic programming problems with dimension higher than two or three, some other device has to be employed. In section 5, several computational techniques which have appeared in the hydrologic literature and which afford an improvement over discrete dynamic programming are discussed and compared.

Water resource problems have served as a stimulus to the development of dynamic programming itself, and many of the studies to be surveyed have attracted the attention of workers outside hydrology. Water allocation problems suggested by Warren Hall are to be found in *Bellman*'s [1957, p. 144] foundational book. A reservoir operations problem figures prominently in *Larson*'s [1968] exposition of the state increment dynamic programming technique. Moreover, a reservoir control problem is among the few selected applications studied in the recent treatise on stochastic dynamic programming by *Dynkin and Yushkevich* [1979]. An unmistakable conclusion is that water resource problems serve as an excellent impetus and laboratory for dynamic programming developments; conversely, progress in making dynamic programming applications in water resources economically viable depends on further advances in theoretical and numerical aspects of dynamic programming. At the present time the influence of dynamic programming on water resource practice is modest. Attempts will be made in this survey to point out where further mathematical or modelling efforts are needed.

2. NOTATION, PERSPECTIVES, AND PRELIMINARIES

At the outset of his seminal and celebrated book *Dynamic Programming*, *Bellman* [1957] defined 'dynamic programming' to be 'the theory of multistage decision processes.' Since that time, some workers have come to associate the name dynamic programming with a particular approach we will later identify as 'discrete dynamic programming.' However, one theme of the present survey is that progress with

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respect to important water resource problems demands that we keep in mind Bellman's original broad conception.

A deterministic multistage decision process, or (discrete time) 'optimal control problem,' as we will hereafter call it, is characterized by an 'initial state' x_1 and functions f_t and L_t as follows: The functions f_t determine relations between 'controls' $\{u_t\}$ and 'states' $\{x_t\}$ according to

$$x_{t+1} = f_t(x_t, u_t) \qquad 1 \le t \le N \tag{1}$$

The x_t's are presumed to be real n-tuples and the u_t's are m dimensional. The set of decision stages, i.e., the domain for the index t in (1) are the first N positive integers, or the set of all positive integers for an infinite horizon process. The function $f_t(x, u)$ will be referred to as the 'dynamics,' and $L_t(x, u)$ as the 'single-stage loss' function.

With any sequence $\mathbf{u} = \{u_t\}$ of controls (such a sequence being known as a 'policy'), one associates a real-valued objective function defined by

$$J(\mathbf{u}) = \sum_{t=1}^{N} L_t(x_t, u_t) \tag{2}$$

The goal, with respect to an optimal control problem (OCP), is to construct a policy $\mathbf{u}^*$ which minimizes the objective function $J(\mathbf{u})$. Typically, the 'feasible' (allowable) controls are those which satisfy a vector-valued state-stage dependent constraint of the form

$$g_t(x_t, u_t) \le 0 \qquad 1 \le t \le N \tag{3}$$

While the stage index t often refers to 'time,' this is not always the case. For example, in section 3 it variously indexes the number of projects already completed, the number of pollution sources downstream from a certain point on a river, etc. In brief, the modifier 'dynamic' of dynamic programming is somewhat of a misnomer, and recognition of the broader nature of the stage variable can enhance one's modelling capabilities.

A popular dynamic programming procedure for the finite horizon OCP's begins by recursively solving what *Bellman* [1957] calls the 'functional equation,' namely, successively determining the 'optimal return functions' $V_N(x)$, $V_{N-1}(x)$, $\cdots$, $V_1(x)$, by the recursive equation

$$V_t(x) = \min_u \left[L_t(x, u) + V_{t+1}(f_t(x, u))\right] \tag{4}$$

$$t = N, N-1, \cdots, 1$$

where $V_{N+1}(x) \equiv 0$. The minimization is with respect to controls satisfying (3). Let $u_t(x)$ denote a control minimizing the bracketed term in (4). It is readily proven that the policy $\mathbf{u}^*$ determined by $x_1^* = x_1$, and

$$u_t^* = u_t(x_t^*), \; x_{t+1}^* = f_t(x_t^*, u_t^*) \qquad t = 1, \cdots, N \tag{5}$$

is a global minimizer of $J(\mathbf{u})$; i.e., $\mathbf{u}^*$ is a solution of the OCP.

Bellman and Dreyfus [1962] advanced a procedure which has come to be called 'discrete dynamic programming.' Discrete dynamic programming (DDP) is a technique for computer implementation of the above functional equation approach to solving OCP's. In essence, DDP requires that the state and control spaces be discretized by finite sets of vectors, which we will denote by $\{x^{(i)}\}_{i=1}^{I}$ and $\{u^{(j)}\}_{j=1}^{J}$, respectively. For each stage $t = N, N-1, \cdots, 1$ and state $x^{(i)}$, prototypical DDP replaces the functional equation (4) by

$$\hat{V}_t(x^{(i)}) = \min_{1 \le j \le J} \left[L_t(x^{(i)}, u^{(j)}) + \hat{V}_{t+1}(f_t(x^{(i)}))\right] \tag{6}$$

with the starting condition that $\hat{V}_{N+1}(x) \equiv 0$. A difficulty arises because (6) determines values $\hat{V}_{t+1}(x)$ only at the node points $x^{(i)}$, whereas the state $f_t(x^{(i)}, u^{(j)})$ need not be among the discretized states. As *Bellman and Dreyfus* [1962] observe, this difficulty can be confronted by function interpolation and other schemes, although of course errors thereby arise.

The main problem with the DDP approach is the phenomenon which is popularly known as the 'curse of dimensionality.' To illustrate the 'curse,' let us suppose that each coordinate of the state variable must be discretized into 10 levels to achieve satisfactory approximation accuracy. This implies that if there are 12 state variables (i.e., if $n = 12$), then the number I of discretized state nodes will be 10^{12}, an impossibly large number to deal with on present or foreseeable computers. Workers in hydrology and elsewhere concede that the limit of the domain of computational applicability of DDP is restricted to problems having at most four or five state and control variables. Some (e.g., *O'Laoghaire and Himmelblau* [1974] and references cited on page 67 thereof) have extended such limitations to all of dynamic programming. As we will argue in section 5, such extrapolation is unwarranted. What is required for solution of large-scale problems is a combination of analytic, nonlinear programming, and numerical analysis ideas, as well as specialized problem-dependent devices. *Bellman and Dreyfus* [1962, p. 21], having outlined the DDP procedure and its promising features, wrote

> Why it is that with all these advantages we do not have a routine solution to all types of allocation processes will be discussed below. It can safely be said that there is still great need for ingenuity, and that much fascinating research remains to be done before a large number of significant processes can be treated effectively.

During the two decades which have passed since the above words were written, considerable progress has been made. The major techniques such as 'state increment dynamic programming,' 'discrete differential dynamic programming,' and 'differential dynamic programming' which address the computational problem have worked their way into the hydrologic literature and will be described in section 5 of this survey. Our view is that there are potent methods available for discrete, deterministic optimal control problems. This view will be supported in section 5. It suffices to say here that some of the methods to be described do overcome the 'curse of dimensionality' of exponentially increasing computational burden with increase in state dimension.

Virtually every water resource system has some random components. Thus stochastic optimal control problems are of considerable importance to the hydrologist. A 'stochastic optimal control problem' differs from an optimal control problem mainly in that the dynamics function $f_t(x, u)$ of the OCP is replaced, in the stochastic version, by a 'probabilistic transition function' $p_t(y; x, u)$ which for fixed x, t, and u is a probability density function in y. This function is presumed to specify the probability distribution of the random state X_{t+1} if at stage t the state is x and the control u is applied.

In the stochastic case it is sometimes useful to make the single-stage loss function L_t depend explicitly on the successor state x_{t+1}; thus we will write $L_t(x_t, x_{t+1}, u_t)$. (In the

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TABLE 1. Applications to Design and Water Quality Control in Rivers and Sewers

Author(s)	Subject	State	Control	Components of Single-Stage Loss	Basis for Dynamics	Stage
Dracup and Fogarty [1974]	Control of river temperature	Temperatures at discharge sites	Amount of flow sent to cooling tower pond, or bipass pipe	Costs due to cooling and bipassing	Mixing and temperature profile model	Position on river
Labadie et al. [1974]	Control of sewage overflow during rainstorm	Flowthrough and storage of various reservoirs	Storage release and adjustment of weirs	Volume of sewer overflow	Muskingum routing model	Time
Leibman and Lynn [1966]	Maintenance of dissolved oxygen along a river	Dissolved oxygen at various effluent locations	Extent of effluent treatment	Cost of effluent treatment	Streeter-Phelps equation for BOD profiles	Position on river
Mays and Wenzel [1976]	Design of branching sewer system	Elevation at upstream end of pipe	Amount of elevation drop in each section of given isonodal level, and pipe diameter	Cost of sewer installation	Sums of elevation drops	Number of manholes from discharge point
Chang and Yeh [1973]	Maintenance of dissolved oxygen along a river	Dissolved oxygen at various effluent locations	Aeration	Cost of aeration and deviation from oxygen standard	Streeter-Phelps equation for BOD profiles	Position on river

deterministic case, x_{t+1} was uniquely determined by x_t and u_t, and so its appearance as a variable would have been superfluous.) In some instances it is convenient to replace the argument x_{t+1} of L_t by a random variable D, say, having a specified probability density function.

A 'strategy' S is defined to be a sequence of functions which, for each state x and stage t, selects a feasible control, $u_t = S_t(x)$. A strategy S having been selected for a given stochastic optimal control problem, the probability law for the states $X_2, \cdots, X_N$ is uniquely determined, and thus the expectation

$$J(\mathbf{S}) = E[\Sigma_{t=1}^{N} L_t(X_t, X_{t+1}, S_t(X_t))] \tag{7}$$

is well defined. A 'solution' to the stochastic optimal control problem is, of course, a strategy S* which minimizes $J(\mathbf{S})$.

The dynamic programming concept of functional equation is, with minor modification, pertinent to the stochastic OCP setting. In the stochastic case it takes the form

$$V_t(x) = \min_u [\int (L_t(x, y, u) + V_{t+1}(y)) p_t(y; x, u)\, dy] \tag{8}$$

with $V_{N+1}(x) = 0$, for all x. The strategy S* determined by $S_t^*(x) = u^*$ with u^* any minimizer of the right-hand side of (8) is easily proven to be optimal. The optimal return function $V_t(x)$ having been reinterpreted as above, the discrete dynamic programming method (especially (6)) is directly applicable to computing optimal strategies.

The stochastic version of (6) is determined by the terminal condition $\hat{V}_{N+1}(x^{(i)}) = 0$, $1 \le i \le I$, and for $t = N, N-1, \cdots, 1$ and $i = 1, 2, \cdots, I$, by

$$\hat{V}_t(x^{(i)}) = \min_{1 \le j \le J} \left[\sum_{k=1}^{I} [L_t(x^{(i)}, x^{(k)}, u^{(j)}) + \hat{V}_{t+1}(x^{(k)})] P_t(x^{(k)}; x^{(i)}, u^{(j)}) \right] \tag{9}$$

where for fixed $x^{(i)}$ and $u^{(j)}$, P_t is a probability mass function associated with transition function p_t and the discretization scheme used. Computational techniques such as differential dynamic programming, which are so effective for deterministic OCP's, are not directly relevant to the stochastic versions.

In section 5 of this survey, a case will be made that concepts from the literature of integral equations, inventory theory, and diffusion theory have a bearing on certain hydrological problems. Yet in the stochastic case, discrete dynamic programming remains the only approach of any universality. It appears that considerable effort and analytic prowess will be required before optimal strategies will be found to realistic stochastic optimal control problems.

This introduction is undeniably brusque and abridged. For a more complete background and motivation for optimal control and dynamic programming, we suggest choosing from the following book selections: *Bellman and Dreyfus* [1962], *Bertsekas* [1976], *Dreyfus and Law* [1977], *Howard* [1960], *Larson and Casti* [1978], *Nemhauser* [1966], and *Yakowitz* [1969]. Stochastic problems with an infinite number of decision times necessitate delicate mathematical considerations. Books dealing with these subjects include *Dynkin and Yushkevich* [1979], *Hinderer* [1970], and *Bertsekas and Shreve* [1978]. The book edited by *Puterman* [1978] and the many references therein may be of use to readers wishing to acquaint themselves with recent literature on the theoretical aspects of dynamic programming.

3. HYDROLOGIC MODELS

Water Quality Maintenance in Rivers and Sewers

The first three case studies to be described are concerned with water quality maintenance, and the last study in this subsection formulates and solves a design problem of minimum cost installation of a branching sewer system, the inflow, confluence, and discharge locations having already been selected. With only one exception, these works have the common structural feature that the decision stage index t is associated not with time, but with discrete locations along the reach of the river or the sewer system in question. All the process models in this section are deterministic, by virtue of the authors' resolve to work with expected flows.

For the applications in this section, we provide (Table 1) a

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summary of the optimal control problem components; such a tabulation will be given for each of our application groupings.

Dracup and Fogarty [1974] consider the problem of reducing thermal pollution in a section of a river, the heat being the result of using a portion of the river water to cool power plants at various locations. The means for reducing the river temperature are (1) cooling towers and (2) bypass piping, which allows a portion of the heated water to enter downstream after the stream has had some chance to cool, and (3) cooling ponds. The river flow is assumed known, and decision stage t, $t = 1, 2, \cdots, N$, indexes the location of the tth power plant. The loss associated with each stage and decision reflects the cost of cooling for that stage and the maximum temperature in the reach between that stage and the next downstream power plant location. A numerical study, based on data of projects for the Green River in Utah, is presented.

The problem of minimum cost maintenance of a dissolved oxygen level along a portion of a river is considered by *Leibman and Lynn* [1966]. The oxygen deficit arises because of effluent discharges at certain points along the river; control is exerted by way of degree of treatment of this effluent. The minimum level of dissolved oxygen is computed, on the basis of effluent composition and dissolved oxygen at the discharge point, by the Streeter-Phelps model. A numerical computation based on data of the Willamette River in Oregon is presented. This study is structurally very closely allied with the preceding *Dracup and Fogarty* [1974] study, which as we saw also has stages indexing location of release points, states which give the (thermal) pollution level, and control exerted by the level of treatment at the release locations.

The study by *Chang and Yeh* [1973] has also examined the problem of oxygen deficit. The essential departure from the work of *Leibman and Lynn* [1966] is that the former includes a term in the loss function which is proportional to the square difference between a desired standard and actual dissolved oxygen at each river location, whereas Leibman and Lynn had taken the oxygen standard as a constraint. However, Chang and Yeh include as constraints an upper bound on the oxygen deficit as well as the total aeration capacity. A numerical example illustrates their dynamic programming solution procedure.

Mays and Wenzel [1975] study the minimum cost choice of pipe slope for a branching storm sewer system, the locations of confluence, inflows, and discharge being given. The stages of the optimal control problem are determined by the number of manholes lying between a pipe in question and the discharge point (which is unique) of the system. The control decision at each stage t is a vector specifying the elevation changes and sizes of pipe, the state being a specification of the upstream elevations of the pipes at a given stage. A dynamic programming solution for an idealized example is given. *Chow et al.* [1975] employ a similar model for one of the tests of their dynamic programming method.

Labadie et al. [1974] consider the problem of sewer overflow in response to rainstorm stresses. The capability for control arises in that storage tanks can be used to divert a portion of the flow; alternatively, weirs are presumed adjustable so as to change the effective storage of portions of the sewer. The authors document that a number of cities have computer-based telemetering and control facilities in the planning stages or already installed. The study under consideration is particularly oriented to a plan being considered by San Francisco; *Labadie et al.* [1975] supplement and extend this work. The solution procedure employs a decomposition into single-storage regions and then uses dynamic programming to minimize spillage in each of these regions during a rainfall event. The Muskingum flow model is used as basis for the dynamics.

Water Resource Project Development

The studies considered in the present section are, for the most part, phrased in terms of reservoir system development. However, it is evident that, formally, the models and methods are appropriate whenever one has a list of possible projects of any sort, an associated cost-benefit measure, and perhaps some overall limitation of some resource (such as money or water) to be allocated among the projects.

Hall and Buras [1961] is a germinal study for the particular genre of dynamic programming applications considered now. These authors issue a 'call to arms' to bring quantitative methods to bear on water resource development activities. The specific 'sizing' problem they dwell on may be described as follows: There are reservoir sites $t = 1, 2, \cdots, N$ available. One must choose how much water capacity u_t should be allotted to the reservoir at site t. A loss function $L_t(u_t)$ measures the cost and benefit if the tth site has capacity u_t. The objective is to choose a policy $\mathbf{u} = \{u_t\}_{t=1}^{N}$ to minimize

$$J(\mathbf{u}) = \sum_{t=1}^{N} L_t(u_t) \tag{10}$$

subject to the constraint that

$$\sum_{t=1}^{N} u_t \leq Q \qquad u_t \geq 0 \tag{11}$$

where Q represents some capacity constraint on the overall system (e.g., net river water availability). Because of the constraint (11), (10) constitutes a resource allocation problem that cannot be trivially solved by minimizing each function $L_t(u_t)$ independently. This resource allocation problem is indeed of the type *Bellman* [1957] used to motivate dynamic programming. For the OCP model for this problem, one may take the state x_t to be the net capacity already allocated. That is,

$$x_t = \sum_{v=1}^{t-1} u_v$$

The corresponding dynamics function is $f_t(x, u) = x + u$. *Hall and Buras* [1961] show how the decision problem of allocating water at a given reservoir site among various uses (flood control, recreation, irrigation) can also be posed as an OCP. Thus dynamic programming can be employed in the construction of the single-stage loss functions $L_t(u_t)$ themselves. *Meier and Beightler* [1967] adapt the methodology of *Hall and Buras* [1961] to sizing reservoirs arranged in a branching configuration.

Trott and Yeh [1973] provide a different OCP formulation to the reservoir sizing problem. Their approach (which was anticipated by *Hall and Howell* [1963]) is: for any fixed vector of reservoir capacities, solve for the associated optimal operating policy, according to principles to be

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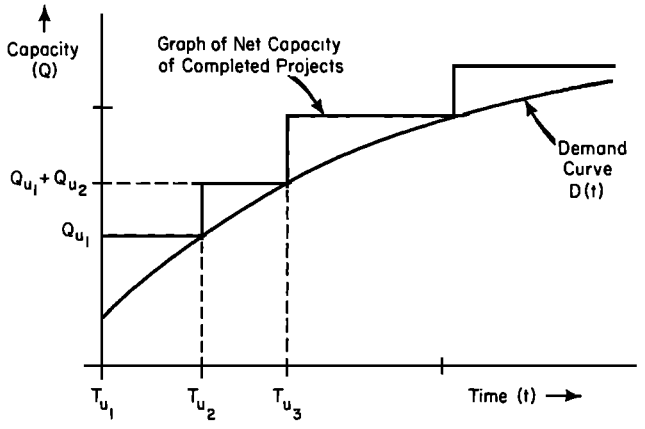


Fig. 1. Capacity plot of an admissible timing sequence.

outlined in section 4 of this survey. Then adjust the hypothetical reservoir capacity vector in order to find the vector that maximizes the net annual benefits. The optimization over capacity vectors is achieved by nonlinear programming methods.

Riordan [1971] studies essentially the same reservoir sizing problem. But he has given further basis for the selection of the loss function L_t, based on economic theory of 'net benefit.' Riordan also discounted future costs, a practice found in several papers mentioned in this section.

A line of inquiry into water resource project timing and sequencing seems to have been initiated by *Butcher et al.* [1969]. The problem differs from the sizing problem discussed above in that here the reservoir sizes Q_i and project costs C_i, $1 \le i \le N$, are presumed specified in advance, and the problem is to find the order and the times at which the projects should be completed so as to minimize the discounted sum of project costs. The first control u_1 designates the project to be completed first; u_2 specifies the second project, etc. A monotonically increasing demand function $D(T)$ is specified, with the interpretation that at all times T, the cumulative capacities of reservoirs already completed up to this time must meet or exceed $D(T)$. Figure 1 shows a demand function and a cumulative capacities plot associated with an admissible project timing sequence $\{T_{u_i}\}$, T_{u_i} being the time of completion of the ith installation.

There are two possible interpretations of what *Butcher et al.* [1969] intend as the state space. One is that the one-dimensional state represents the cumulative capacities of the reservoirs already completed. As *Morin* [1973] observes, this state specification does not satisfy the postulates of the OCP model (as specified in section 2). Specifically, it is possible for two distinct sets of projects to yield the same cumulative capacity. In this case, one cannot tell, at a given state and stage, which projects are still available. As a consequence, the constraint function (3) is not well defined. *Morin* [1973] outlines how to sidestep this problem by ensuring a one-to-one mapping between cumulative capacities and sets of projects. Additionally, he observes connections between project sequencing and scheduling problems and the operations research literature.

Morin and Esogbue [1971] note that the dynamic programming solution, with cumulative capacities as state space, can be carried out efficiently by using the fact that only a limited set of capacity values are possible. *Becker and Yeh* [1974*a*] extend the techniques of *Butcher et al.* [1969] to the task of project expansion for meeting firm water yields. *Moore and Yeh* [1980] augment the model by showing, in the spirit of *Riordan* [1971], how economic principles such as price elasticity, demand functions, and marginal costs can be incorporated into the single-stage loss function. *Morin* [1974] points out that exact solution of a sequencing problem can be computationally expensive if the number of projects is larger than 20. He further describes an inexpensive heuristic scheduling algorithm and gives a sufficient condition that the solution it provides indeed be optimal.

The details of a dynamic programming procedure for the timing and sequencing problem are now presented. It turns out that once the optimal sequence of projects has been found, the timing problem is automatically solved; you delay the completion of each project as long as possible, subject to the demand constraint not being violated. Specifically, if $\mathbf{u} = (u_1, \cdots, u_N)$ is the optimal policy, the time T_{u_i} of completion of the ith installation is given by

$$T_{u_i} = D^{-1}\left(\sum_{t<i} Q_{u_t}\right) \tag{12}$$

D^{-1} being the inverse of the demand function.

Erlenkotter's [1973*a*, *b*] solution to the project sequencing problem of *Butcher et al.* [1969] is as follows: the state x_t at stage t is the set of projects which have been completed. (Such sets can be encoded as real numbers and thereby made to satisfy the OCP postulates of section 2.) Then the optimal return function satisfies the functional equation

$$V_t(x) = \min_{u \notin x} [C_u(1 + r)^{-T(x)} + V_{t+1}(f_t(x, u))] \tag{13}$$

where $T(x) = D^{-1}(\Sigma_{i \in x} Q_i)$ and the transition function is given by the union operation $f_t(x, u) = x \cup \{u\}$. As in the work of *Butcher et al.* [1969], C_u and O_u are the cost and capacity of the project u, respectively, and r is the discount rate. (Actually, *Erlenkotter*'s [1973*a*, *b*] solution is more general in that, as he notes, the cost C can be made to depend on x as well as u.)

Kaplan and Haimes [1975] called upon techniques of *Butcher et al.* [1969] to obtain a dynamic programming solution to the problem of scheduling expansions of a fixed wastewater treatment facility. The control at each decision time t is to be chosen from a fixed set U of possible plant expansion increments. The resulting OCP bears some similarity to the project scheduling problem but is actually much easier in that U remains the same (any particular increment being available for use at any number of stages).

The final study described in this subsection concerns capacity expansion of a firm water system (such as a network of artesian wells). It is presumed that the firm water system supplements a single reservoir whose storage is subject to random fluctuations. *Bogle and O'Sullivan* [1979*b*] presume that at yearly intervals one can decide, on the basis of the amount of water q_{t-1} stored in the reservoir and the current capacity u_{t-1} of the supplementary source, whether or not to increase this supplementary source (the 'plant') to a new capacity. The reservoir storage q_t is presumed to be a finite Markov chain with known transition matrix and initial distribution. The state may be taken as $x_t = (q_{t-1}, u_{t-1})$. The single-stage loss function for the Bogle and O'Sullivan model can be represented as

$$L_t(x_t, u_t) = C(u_{t-1}, u_t) + \Sigma_v D_t(v, x_t, u_t)P(v|x_t) \tag{14}$$

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TABLE 2. Applications to Project Development

Author(s)	Subject	State	Control	Components of Single-Stage Loss	Basis for Transition Function	Stage
Hall and Buras [1961] *Hall* [1961] *Riordan* [1971]	Allocation of capacity to various reservoir sites; allocation of water to various uses at each site	Capacity of reservoirs already completed	Reservoir capacity at given site	Costs of construction and benefits to users	Net sum of reservoir capacities	Position from reservoir farthest downstream
Butcher et al. [1969] *Morin and Esogbue* [1971] *Morin* [1973] *Becker and Yeh* [1974*a*]	Timing and sequencing of water supply projects	Index of projects already finished and associated net capacity	Project to be completed next	Costs of construction and benefits to users	Set representations of projects completed	Number of already completed projects
Bogle and O'Sullivan [1979*b*]	Expansion of firm water supply capacity in face of random reservoir supply	Size of current plant capacity, and reservoir storage	Size of expansion	Costs of expansion, water deficit	Sum of capacities, mass balance	Time (annual)

where $C(u, u')$ measures the cost of increasing the plant capacity from u to u', and $D_t(v, x_t, u_t)$ gives the expected operations cost over 1 year if the inflow during the year is v, the water stored in the reservoir at the beginning of the year is q_{t-1}, and the plant has capacity u_t. The conditional probability $P(v|x_t)$ of inflow is determined by the Markov transition matrix. In their numerical studies of a representative problem, *Bogle and O'Sullivan* [1979*b*] found that the two-dimensional state-space stochastic OCP problem thus engendered was too complex for discrete dynamic programming solution and proposed an alternative heuristic scheme for approximating the optimal strategy.

Table 2 summarizes applications to project development.

Irrigation Management

The optimal control problems (OCP's) engendered by the question of how much to irrigate, in the face of a limited water supply, are more subtle and complex than those described in the two preceding groups of models. The dynamic programming methodology for the irrigation control process problems is correspondingly a bit more delicate. In the first of the studies to be considered here, it takes some thought to see how to phrase the problem as a classical OCP, and the remaining studies employ stochastic OCP's. Some of the suggested means for computational solution of these models are reasonably sophisticated, calling upon concepts of inventory and Markov chain theories. We shall defer more than passing comment on the computational issues to section 5.

The earliest applications of dynamic programming to the irrigation problem to come to our attention are the closely related papers of *Flinn and Musgrave* [1967] and *Hall and Butcher* [1968]. In the latter study it is theorized that the effect on crop yield of soil moisture deficiency depends critically on the stage of the growing season during which the deficiency occurs. For a fixed crop acreage, a quantity $Y_{\max}$ is defined to be the crop yield if soil moisture conditions are ideal during all N growth stages. Let $w_1, \cdots, w_N$ be a sequence of soil moisture values, where the subscript indexes the N growth periods of the season. Then it is presumed that the actual yield is given by

$$y = \left[\prod_{t=1}^{N} a_t(w_t)\right] Y_{\max} \tag{15}$$

the a_t's being known functions of the soil moisture variables w_t. The first OCP considered is that of allocating a limited (by q_0) amount of water among the N growth stages to maximize the yield y. For this problem the state variables are q_t and w_t, the reservoir content and soil moisture at growth stage t, respectively. The dynamics are determined by

$$\begin{pmatrix} q_{t+1} \\ w_{t+1} \end{pmatrix} = f_t[(q_t, w_t), u_t] = \begin{pmatrix} q_t - u_t \\ w_t + \eta u_t - e_t \end{pmatrix} \tag{16}$$

where u_t, the control variable, indicates the amount of water used for irrigation at the tth stage, η is a fixed 'efficiency' constant determining the soil moisture response to irrigation, and e_t is a known quantity summarizing the effects of water loss due to evaporation, transpiration, and deep percolation. The single-stage loss function is

$$L_t(x_t, u_t) = -\ln a_t(w_t) \tag{17}$$

and the choice of controls is constrained by

$$0 \le u_t \le q_t \qquad w_1 \le w_{t+1} \le w_2 \tag{18}$$

where w_1 and w_2 are bounds for allowable moisture. One can check that any policy $\{u_t\}$ minimizing $\Sigma - \ln [a_t(w_t)]$ will maximize the yield $[\Pi a_t(w_t)]Y_{\max}$.

The irrigation OCP was then augmented so as to reflect the cost of irrigation applications. Specifically, the overall loss was taken to be

$$J(\mathbf{u}) = \sum_{t=1}^{N} c_t(u_t) - \prod_{t=1}^{N} a_t(w_t) Y_{\max} \tag{19}$$

To convert this optimization problem to one within the framework of our OCP construct, it seems necessary to add a state variable which keeps track of the growth factor. Thus if a third state variable α is defined so that at each decision time t

$$\alpha_t = \prod_{j=1}^{t} a_j(w_j) \tag{20}$$

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then another coordinate is added to the dynamics law (16), to wit:

$$\alpha_{t+1} = a_t(w_t)\alpha_t \tag{21}$$

With the state variable now given by the triple $x = (q, w, \alpha)$, the single-stage loss function appropriate to (19) becomes

$$L_t(x_t, u_t) = c_t(u_t) \qquad t < N$$

$$L_t(x_t, u_t) = c_N(u_N) + Y_{\max}\alpha_N \qquad t = N$$

Dudley et al. [1971] provide a stochastic irrigation model which has many of the components of the *Hall and Butcher* [1968] study just considered. The central difference in the dynamical system is that the component e_t, summarizing effects of transpiration, etc., in (16) is now random. The effect of this assumption is that the state variable w_{t+1} is determined by a probability density function $p_t(w_{t+1}; w_t, u_t)$.The single-stage loss function depends only on the state $x = (q, w)$, where q measures the reservoir content and w the soil moisture, and control u, which is the irrigation level. *Dudley et al.* [1971] used a straightforward discrete stochastic dynamic programming algorithm to obtain a numerical solution to an irrigation problem based on data from Inverell, New South Wales. Among the approximations introduced to simplify the computations is that the irrigation strategy be restricted to the rule: Irrigate at time t if and only if the moisture state coordinate w_t is less than some fixed prescribed soil level. Such strategies are popular in the inventory literature. A more direct connection between irrigational OCP's and inventory control will be established when we describe the work of *Fogel et al.* [1976].

Dudley and Burt [1973] imbed the irrigation operations problems within the system design task of choosing (1) the reservoir capacity, (2) the acreage to be developed for agricultural purposes, and (3) the distribution capacity of the irrigation system. The master plan for attacking this systems design problem is to find the optimal operation rule for any given design parameters and to choose the design parameters to be those under which the optimal operation policy gives minimum loss. This plan harks back to concepts in the work of *Trott and Yeh* [1973] mentioned in the preceding section. The bulk of the study by Dudley and Burt is devoted to solving the operation problem itself. To the basic state variables q and w of the preceding two papers, *Dudley and Burt* [1973] add 'ϕ' and 'a.' The variable a_t denotes the acreage available for irrigation, and ϕ_t is a 'crop production function' which depends on climatic and cultural variables as well as soil moisture. The sum $\Sigma_t\phi_t$ of the crop production function values over the various stages of a season is a measure of the yield of the crop.

The two control variable coordinates are (1) the level of irrigation (as in the previous two papers) and (2) the acreage to be irrigated. The last control variable arises from an option allowed by *Dudley and Burt* [1973] that portions of the planted crop may be abandoned to wilt if a water shortage develops.

During 'intrayear' periods the loss function depends on the irrigation costs, as determined by the irrigation level and the acreage of land being farmed. At times which represent the end of the growing season, the cost has a term equal to $\Sigma\phi_t$ which gives the sum of the production function values over the growing season and represents the net harvest value of the crop. As in the preceding work, the state transition is presumed to be random, thus engendering a stochastic optimal control problem. For computational convenience the soil moisture variable is ignored; even so, according to the authors, the computational demands are excessive. An interesting computational approach is therefore introduced. *Dudley and Burt* [1973] argue (undoubtedly correctly) that as the number N of decision stages becomes large, the optimal decision at any given state should depend on time only through the period of the growing season. That is, the optimal strategy is, in the words of *Hinderer* [1970], for example, 'stationary'; it depends only on state, not time. The important consequence of having stationary strategies $S_t(x) = S(x)$ is that the state variable process must be a Markov chain. This enables one to employ some powerful analytic machinery such as described by *Feller* [1968] to get insights about the process. The insight *Dudley and Burt* [1973] exploit is that once the state space is discretized, one can find asymptotic probability of occurrence of the various states by solving a linear equation. From these asymptotic probabilities, the expected loss under any stationary strategy is readily calculated. One then sifts among stationary strategies to find the one minimizing this expected loss. *Bogle and O'Sullivan* [1979*a*, *b*] also adopt this heuristic technique.

Dudley et al. [1976] have applied the methodology of *Dudley and Burt* [1973] to a related irrigation problem motivated by data associated with the Tiber Dam in Montana. In the work of *Dudley et al.* [1976], the state variables are (1) reservoir content and (2) the acreage of winter wheat existing in the spring. The decision variable is acreage of winter wheat to plant and the level of irrigation. The proposed method of solution is essentially that of *Dudley and Burt* [1973] just described. *Rhenals and Bras* [1981] have provided an irrigation model which accounts for random evapotranspiration as well as soil moisture and water reserve, as in the works just mentioned. Care was given to construction of realistic dynamics based on current physical understanding of irrigation phenomena. A solution to a numerical problem was obtained through discrete stochastic dynamic programming.

Fogel et al. [1976] establish a direct link between irrigation management and inventory theory. In a simple and standard form of the inventory problem, one is concerned with a 'set up cost' c_1 which is incurred whenever an order of any size is placed, a unit cost c_2 per item ordered, a 'unit holding cost' c_3 reflecting the expense of storing a unit of merchandise from one decision period to the next, and a 'unit shortage cost' c_4 reflecting the expense, per unit, of unfulfilled demand. The interpretation is that at the beginning of the tth decision stage the inventory on hand is x_t. The decision maker orders an additional quantity u_t, at a cost of $c_1 + c_2u_t$ (or 0, if $u_t = 0$). A random demand D_t is observed, and if there is a shortage (i.e., if demand D_t exceeds $x_t + u_t$), then one incurs a shortage cost $c_4[D_t - (x_t + u_t)]$. Otherwise, there is a storage cost of $c_3(x_t + u_t - D_t)$.

In summary, this inventory model presumes that the single-stage loss is given by

$$L_t(x_t, D_t, u_t) = c_1l(u_t) + c_2u_t + c_3(x_t + u_t - D_t)^+ + c_4(D_t - x_t - u_t)^+ \tag{22}$$

In (22), $l(a)$ represents the unit step function which is one whenever $a > 0$ and 0 otherwise and $(a)^+ = \max\{a, 0\}$.

As *Hadley and Whitin* [1963] document, some of the earliest analyses with dynamic programming functional equation techniques were motivated by the inventory prob-

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TABLE 3. Application to Irrigation Management

Author(s)	Subject	State	Control	Components of Single-Stage Loss	Basis for Dynamics
Hall and Butcher [1968]	Irrigation level and timing, in face of limited water supply	Soil moisture, water supply, and crop measurement function	Irrigation level	Irrigation cost and net crop yield	Water mass balance, and biological growth rate as function of soil moisture
Dudley et al. [1971]	Irrigation level and timing, with limited water, and stochastic crop water needs	Same as above	Same as above	Not made explicit	Same as above
Dudley and Burt [1973]	Same as above	Same as above as well as acreage being cultivated	Irrigation level and acreage to be irrigated	Irrigation cost and net crop yield	Same as above
Fogel et al. [1976]	Irrigation level and timing, limitless water supply	Soil moisture	Irrigation level	Irrigation cost, crop damage due to insufficient moisture	Same as above

lem; several of these works predate Bellman's dynamic programming publications. Dynamic programming methodology is the medium for proving the standard result [*Bellman,* 1957, chapter 5] that the discounted, nonterminating OCP with the single-stage loss function (22) has an optimal strategy of the form (for some fixed constant α)

$$\begin{aligned} u(t) &= Q - x_t \qquad x_t \leq \alpha \\ u(t) &= 0 \qquad\quad \text{otherwise} \end{aligned} \tag{23}$$

By 'discounted' OCP, we mean one in which the overall loss function $J(\mathbf{S})$ weighs future losses at stage t by the discount factor a^t. That is, $J(\mathbf{S}) = \Sigma_t a^t E[L_t(x_t, D_t, S_t(x_t))]$.

In *Fogel et al.* [1976], the following identification of irrigation quantities with inventory constructs is made:

- c_1 cost of an irrigation application (due to laying pipes, clearing trenches, etc.);
- c_2 expense of a unit of irrigation water;
- c_3 damage expense due to excess soil moisture, water loss due to evaporation and percolation;
- c_4 damage due to crop reduction resulting from insufficient water;
- D_t water consumption during tth decision interval of the irrigated field through all effects (transpiration, percolation, etc.);
- u_t quantity of water applied at tth decision time;
- x_t amount of soil water on hand prior to tth irrigation.

It is to be recognized that the assumption that crop damage due to insufficient moisture varies linearly with moisture itself is patently questionable. In this regard, it is worth noting that strategies of the form (23) are known [e.g., *Hadley and Whitin,* 1963] to be optimal for a larger class of cost functions.

An important difference between the irrigation decision problems considered by the other authors we have cited and that of *Fogel et al.* [1976] is that the latter paper makes no assumption concerning limitations in water supply; the only costs considered are those involved in the expense of irrigation applications.

The reader is referred to *Matanga and Mariño* [1979] for a dynamic programming irrigation model in which soil salinity and leaching effects are considered.

Table 3 summarizes the OCP irrigation management models.

Other Applications

In this subsection we consider two application areas which have not attracted many studies but nevertheless seem worthy of comment. Of course, yet other water resource dynamic programming applications exist. Some models are to be found scattered among the references in the 'Systems Analysis' section of the literature review issues of the *Water Pollution Control Federation Journal.*

The first of the areas we discuss is that of aqueduct design [*Hall,* 1961]. The model and technique to be described would appear to be applicable to any pipe or conduit layout problem and to the design of irrigation systems, in general.

Hall [1961] presumes that the route and N outflow points of the aqueduct have already been determined. The problem (see Figure 2) is to select pipe capacity of the individual links so as to optimize the tradeoff between the value of water delivered to the various turnouts and the construction cost of the aqueduct system. Thus $v_i(u_i)$ represents the benefit to the ith geographical district of making water at flow rate u_i available to that district. Originally, the optimization problem was stated to be that of selecting pipe capacities so as to minimize

$$J(\mathbf{u}) = -\sum_{i=1}^{N} v_i(u_i) + C(u_1, \cdots, u_N) \tag{24}$$

subject to the constraint that

$$Q \geq u_i \geq 0 \tag{25}$$

The parameter Q in (25) may be viewed as a bound on the capacity of the aqueduct water source. The function C in (24)

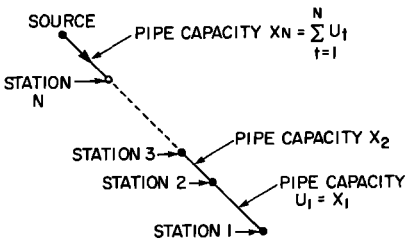


Fig. 2. Illustration for the aqueduct capacity problem.

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gives the construction cost of an aqueduct system having the delivery capacities displayed in its argument.

It turns out that unless the cost function is presumed to have some restrictive mathematical structure, the task of minimizing $J(\mathbf{u})$ is not an optimal control problem, but rather a nonlinear programming problem. The assumption ultimately made by Hall is that

$$C(u_1, \cdots, u_N) = \sum_{t=1}^{N+1} c_t(x_t) \tag{26}$$

where $c_t(x_t)$ is the cost of constructing an aqueduct segment between locations $t+1$ and t capable of handling a flow of x_t units. Location N can be viewed as the aqueduct head and $x_t = \Sigma_{j<t} u_j$. By virtue of location 1 being at the tail of the system, $x_1 = 0$. With this specification, the OCP model can be obtained by taking the net flow to all turnouts $j, j < t$ as the state variable x_t for stage t and the single-stage loss function consistent with (26) is

$$L_t(x_t, u_t) = -v_t(u_t) + c_t(x_t + u_t) \tag{27}$$

The dynamics are given by

$$x_{t+1} = f_t(x_t, u_t) = x_t + u_t \tag{28}$$

The control at each decision stage t is the flow rate u_t of water supplied to the tth turnout. This completes the formulation of the OCP model. *Hall* [1961] obtained a numerical solution to a two-stage problem by a variant of the basic discrete dynamic programming algorithm described in section 2 of this survey. *Chow et al.* [1975] used this aqueduct design task as a test problem for their computational method. *Sathaye and Hall* [1976*a*, *b*] extended the preceding analysis to account for seasonal fluctuations in the water demand and for the possibility of adding a reservoir at some intermediate point along the aqueduct.

Buras [1972] offered a number of OCP models for various water resource activities. He has also described the aqueduct design problem mentioned above and described an extension of the problem so that the actual route of the aqueduct can be incorporated into the design problem.

Perhaps the most sophisticated and thorough case study provided by *Buras* [1972] (following *Buras* [1963]) concerns a problem of conjunctive use of an aquifer and a reservoir to trade off cost of water delivery with costs associated with water having high concentration of chlorine ions (a condition which we will refer to as 'salinity'). The configuration for the conjunctive management problem is illustrated in Figure 3. Our exposition ignores spillage and other details of the Buras

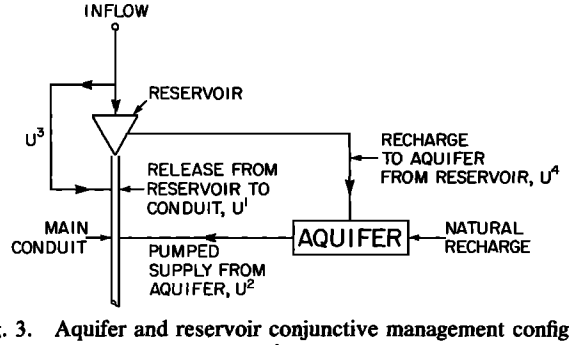


Fig. 3. Aquifer and reservoir conjunctive management configuration.

model. The ultimate sources of supply for the main conduit are as follows: (1) water from the reservoir (Lake Kinneret, in Israel), which has a high salinity level, (2) water from the aquifer, which is low in salinity but incurs extra pumping expense, (3) inflow water, which is of relatively low salinity, diverted directly to the conduit, and (4) water released directly from the reservoir into the aquifer. (The benefit is that it may reduce pumping costs without unduly raising the aquifer salinity level.)

Toward constructing an OCP model, the state coordinates are selected to be $x_t = (q_t, g_t, s_t)$, where q_t is the water stored in the reservoir at decision stage t, g_t is the water content of the aquifer, and s_t is the salinity of the water in the reservoir. The control vector is $u_t = (u_t^{(1)}, u_t^{(2)}, u_t^{(3)}, u_t^{(4)})$, where each coordinate represents the quantity of water of the corresponding above-listed category to be released during the decision stage t. The states evolve stochastically because the inflows to the reservoirs and aquifer are presumed random, with known distribution. The law for state coordinate q is determined by

$$q_{t+1} = q_t - u^{(1)} + Y_t - u^{(3)} - u^{(4)} \tag{29}$$

where Y_t is the random reservoir inflow during the tth decision stage. Similarly,

$$g_{t+1} = g_t - u_t^{(2)} + u_t^{(4)} + Z_t \tag{30}$$

where Z_t is the random aquifer recharge. By mass balance, and assuming negligible salinity in the aquifer and reservoir inflow, one can find the salt concentration transition by finding the salt weights before and after release and dividing the latter by the new reservoir volume. Any source of additional salt can be accounted for in this process. Thus the state transition law has been determined.

Buras [1972] selected as single-stage loss function the function

$$L_t(x_t, u_t) = \alpha(u_t^{(1)}) + \beta u_t^{(3)} + u_t^{(1)} c(s_t) + \delta(u_t^{(2)}, g_t) \tag{31}$$

where $\alpha(u^{(1)})$ and $\delta(u^{(2)}, g)$ reflect the pumping cost from the reservoir and aquifer, respectively, into the main conduit, and $\beta u_t^{(3)}$ is the cost of diverting inflow water directly to the conduit. The term $u^{(1)}c(s_t)$ reflects the damage from using reservoir water at salinity level s_t. The overall expected loss function $J(\mathbf{S})$ determined by the single-stage losses L_t, as defined in (31), was minimized (subject to constraints due to pumping capacities and user water demands) by discrete stochastic dynamic programming. The model presumed that the decision stages were at monthly intervals. Buras appealed to *Bellman* [1957] to conclude that for long enough decision horizon, the optimal strategy should depend on time only through season.

Bogle and O'Sullivan [1979*a*] also consider a problem in conjunctive use of a reservoir and an 'alternative source.' The physical nature of the alternative source is not material to the method. The reservoir water is considered to be free, and the water supplied by the alternative source has a fixed cost per unit delivered. It is assumed that the alternative source must meet the shortfall, if possible, between the given (deterministic) demand and the amount of water released at each decision stage from the reservoir. However, the capacity of the alternative source is limited; if the demand cannot be met, the source delivers at maximum

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TABLE 4. Miscellaneous Applications

Author(s)	Subject	State	Control	Components of Single-Stage Loss	Basis for Dynamics	Stage
Hall [1961]	Sizing of aqueduct conduits	Downstream aqueduct capacity	Size of pipe between adjoining demand points	Benefit of prescribed water capacity, cost of piping	Mass balance	Turnout location
Buras [1972]	Conjunctive management of an aquifer and reservoir in face of salinity	Storage level in aquifer, in reservoir, and salinity level of reservoir	Rate of pumping from aquifer, release rate from reservoir, rate of diversion from reservoir inflow	Pumping costs, penalty for salinity level in water delivered	Water and salt mass balance	Time
Bogle and O'Sullivan [1979*b*]	Conjunctive management of a reservoir and alternate water supply	Storage level in reservoir	Reservoir release rate	Cost of shortfall in water supply, low level of reservoir	Mass balance	Time

capacity. The motivation for deciding on such a restrictive strategy for the alternative source is that it reduces the dimension of the control vector to one variable, namely, the amount to be released from the reservoir. The one system state is the reservoir storage. The single-stage loss function reflects costs due to failure to meet the water demand and undesirability of low storage levels in the reservoir. The reservoir inflow is presumed to be random.

An interesting feature of the solution procedure advocated by *Bogle and O'Sullivan* [1979*a*] is that they restrict the possible set of strategies to a small number of release rules based on 'engineering practice and common sense.' As a result of this restriction, the rule ultimately chosen will in all likelihood be suboptimal, but the choice is made so that a functional equation decomposition technique can be employed. The computational complexity at each stage is vastly reduced because of the limited number of control values allowed by the release rules.

Table 4 summarizes these miscellaneous OCP models.

4. RESERVOIR OPERATIONS

Deterministic Case

It seems a curious state of affairs that the earliest appearances of stochastic dynamic programming algorithms in reservoir operations [*Masse,* 1946; *Little,* 1955] precede applications of deterministic dynamic programming by over a decade. The reason for astonishment is that the stochastic problem is the more subtle and computationally troublesome case.

We were able to trace published studies of deterministic reservoir applications only as far back as *Young* [1967], who studied a finite horizon, single-reservoir operation problem. (It should be observed that ideas in the work of *Hall and Howell* [1963], *Hall* [1964], and *Hall and Roefs* [1966] predate Young's studies.)

In *Young*'s [1967] model the control variable u_t represents the average release of water (i.e., flow rate) during the tth decision period, and the state variable x_t gives the reservoir water volume at the beginning of the tth period; C denotes the reservoir capacity. The OCP dynamics are specified by a mass balance equation

$$x_{t+1} = \min \{C, x_t + i_t - u_t\} \tag{32}$$

where i_t represents the inflow volume which, in the deterministic case, must be presumed known. Young's dynamics model (32) and its multivariable counterpart, or some close variation thereof, have been adopted by most of the other studies reported in this section.

There are two viewpoints forwarded in the literature to support the critical assumption that the inflows are known. One viewpoint is that some rivers are regular enough that their flows are well represented by their expectations. Under this hypothesis, the inflows $\{i_t\}$ are taken to be the expected inflows during the portion of the annual cycle corresponding to the tth decision period. The other justification of determinism is that insight is to be gained by retrospectively seeing what the optimum policy would have been if only we had known what the river inflows were going to be. Within this framework, one takes $\{i_t\}$ to be an observed historical record.

As single-stage loss, *Young* [1967] takes $L_t(x, u)$ to be a function which varies only with release u. This assumption would seem adequate if the sole purpose of the reservoir is to supply water and/or hydroelectric power.

Having clearly formulated the reservoir operations problem as an OCP, Young sets up a dynamic programming–type functional equation for its solution. However, for his numerical example, he adopts a heuristic variant in which the strategy is constrained to be a linear function of the state and estimated future state, the inflows now being presumed random. The linear parameters of the heuristic strategy are calibrated by regression analysis.

An interesting aspect of Young's study is that he uses calculus of variations methods to obtain an analytic solution to a continuous-time version of the deterministic reservoir operations problem. This tack which, to our knowledge, has not been taken up since, would seem to have some merit. For continuous OCP's, like differential equations, are frequently much easier to solve numerically than their discrete time counterparts. The reason for this is that through devices such as 'Pontryagin's maximum principle' or 'differential dynamic programming,' one can frequently convert the OCP into a well-behaved two-point boundary value ordinary differential equation problem. Such problems can often be accurately and routinely solved by the 'shooting' method or alternatively 'quasi-linearization.' Chapters 4 to 6 of *Kirk* [1970] provide an introduction to these ideas. Chapters 2 and

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3 of *Jacobson and Mayne* [1970] give an alternate differential equation transcription.

Hall et al. [1968] present specific ideas for determining the single-stage loss function. In particular, they define

$$L_t(x_t, u_t) = -[P_t u_t + P_{ct}E_{ct}(x_t, u_t) + P_{nt}E_{nt}(x_t, u_t)] \quad (33)$$

where P_t, P_{ct}, and P_{nt} are scheduled unit prices of released water and power supplied during peak and nonpeak hours, respectively, during the tth decision time. The functions E_{ct} and E_{nt} representing energy delivered during peak and nonpeak hours are presumed nonlinear. The authors report a computational study involving records and specifications associated with Shasta Dam in California. Note that in contrast to *Young* [1967], *Hall et al.* [1968] have allowed that the single-stage loss function can depend on state as well as control. In principle, there is no computational disadvantage to this generalization since, as can be seen by inspection of (4), the optimal return function and the strategy will depend solely on state, even if the loss does not. A modelling advantage of the generalization is that now the single-stage loss can be made to reflect desiderata associated with flood protection and recreational uses of the reservoir. In this regard, *Hall et al.* [1969*a*, *b*] modify the work just described to allow incorporation of additional factors (e.g., firm water and on-peak capacity constraints, energy pricing and flood control desiderata, etc.). Also, they outline improvements in the optimization methodology.

Roefs and Bodin [1970] extend the basic model to the multireservoir case. Such multireservoir models have vector-valued states with the jth coordinate, say, $x_t(j)$, giving the content of reservoir j at time t. The control vector is similarly indexed. The dynamics for the multireservoir model has the form

$$x_{t+1}(j) = x_t(j) + i_t(j) - u_t(j) + \Sigma_{k\in K(j)} u_t(k) \qquad 1 \le j \le n \quad (34)$$

where $i_t(j)$ is the natural inflow into reservoir j during decision time t and $K(j)$ is the set of indices for the reservoirs whose releases flow directly (not by the way of other reservoirs) into reservoir j. Thus with respect to Figure 5, $K(7) = \{1, 4, 5, 6\}$.

Heidari et al. [1971] consider a prototypical four-reservoir problem (illustrated in Figure 4) which is probably beyond the scope of discrete dynamic programming because of the curse of dimensionality. They propose a computational scheme they call 'discrete differential dynamic programming' and give a detailed account of their numerical solution. This same problem is also solved in the work of *Larson* [1968] to illustrate his computational procedure, 'incremental dynamic programming.' In fact, this four-reservoir problem has become somewhat of a benchmark test problem; *Nopmongcol and Askew* [1976] use it as a test of the technique 'multilevel incremental dynamic programming,' and in the work of *Chow et al.* [1975] it serves as a test for discrete differential dynamic programming. *Murray and Yakowitz* [1979] have used this four-reservoir configuration to illustrate the efficiency of their 'constrained differential dynamic programming' algorithm. They also enlarged the problem to a 10-reservoir case (Figure 5) and computed the optimal policy. Details of dynamic programming methods mentioned above will be revealed in section 5 of this survey.

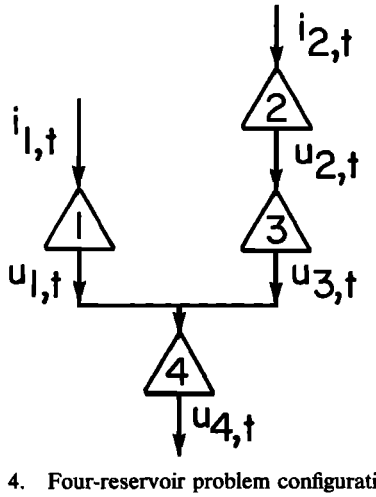


Fig. 4. Four-reservoir problem configuration.

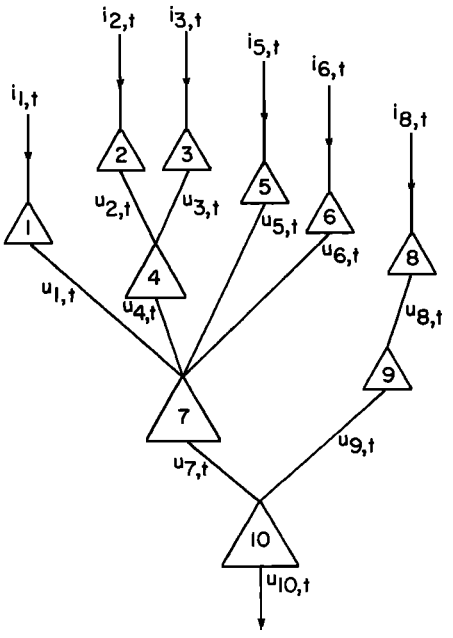


Fig. 5. Ten-reservoir problem configuration.

We have noted in the section 'water resource project development' that *Trott and Yeh* [1973] recognize that the reservoir operations problem must be addressed in the course of reservoir system design. Toward illustrating their procedures for the operations problem, Trott and Yeh apply the incremental dynamic programming method to an OCP model for the operation of a six-reservoir system in California. We comment on two aspects of their study. First, they use the 'forward' dynamic programming functional equation, which differs from (4) in that it satisfies the forward recursive relation, $V_1(x) \equiv 0$, and

$$V_{t+1}(x_{t+1}) = \min_{x_t} \left\{ \min_{u_t \in U(x_t)} [L_t(x_t, u_t) + V_t(x_t)] \right\} \quad (35)$$

where $U(x_t)$ is the set of admissible controls u such that $f_t(x_t, u) = x_{t+1}$. One may interpret $V_{t+1}(x_{t+1})$ as the minimum cost associated with passing from initial state x_1 to state x_{t+1} at stage $t + 1$. Let $u_t^*(x_{t+1})$ be a minimizing control for (35) and $x_t^*(x_{t+1})$ the associated x_t value. The optimal policy is constructed as follows. Let x_{N+1}^* be any state which minimizes $V_{N+1}(x_{N+1})$. Then for $t = 0, \cdots, 1$, successively

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TABLE 5. Applications to Deterministic Reservoir Operations

Author(s)	Subject	State	Control	Components of Single-Stage Loss	Basis for Dynamics
Young [1967] *Hall et al.* [1968]	Operation of a single reservoir	Reservoir storage (volume)	Reservoir release	Not made explicit	Mass balance
Roefs and Bodin [1970] *Heidari et al.* [1971] *Trott and Yeh* [1973]	Operation of a system of several reservoirs	Storage in each reservoir	Release from each reservoir	Negative of profits for water delivered and power generated	Mass balance
Becker and Yeh [1974*a*]	Operation of a system of several reservoirs	Total potential energy in reservoir system	Change in potential energy	Benefits electricity generated, potential energy of stored water	Mass balance

define $u_t^* = u_t^*(x_{t+1}^*)$ and $x_t^* = x_t^*(x_{t+1}^*)$. The policy $\{u_t^*\}$ is readily demonstrated to be optimal. An advantage 'forward' dynamic programming holds, for certain classes of OCP's, over the more conventional 'backward' dynamic programming procedure ((14) and (15)) is that unreachable states can be conveniently ignored: One need evaluate V_{t+1} only at values x_{t+1} for which transition from some reachable x_t is possible. Further discussion of forward dynamic programming is given by *Dreyfus and Law* [1977, p. 10].

The other point concerning *Trott and Yeh* [1973] which is to be noted is that in their numerical study, they use the OCP objective function

$$J(\mathbf{u}) = \max_t \{L_t(x_t, u_t)\} \tag{36}$$

in contrast to the additive objective function (2). It is known [e.g., *Larson and Casti,* 1978, pp. 17–18] that

$$V_t(x) = \min_u \{\max [L_t(x, u), V_{t+1}(f_t(x, u))]\} \tag{37}$$

is a valid dynamic programming functional equation for (36), provided L_t is nonnegative for all arguments.

Becker and Yeh [1974*b*], *Becker et al.* [1976], and *Yeh and Becker* [1979] have described OCP's in which linear programming problems play an integral role. These studies are all directed toward finding operating policies for the Central Valley Project in California. We examine some of the details of *Becker and Yeh* [1974*b*]. For each decision time t, a $2n$ variable linear programming problem is generated: Find values R_t^k and $R_t'^k (1 \le k \le n)$, which minimizes

$$\sum_{k=1}^{n} (c_t^k R_t^k + C_t'^k R_t'^k) \tag{38}$$

subject to certain linear constraints which include on-peak energy generation levels. The objective function (38) reflects the loss of the stored, potential energy of water. Thus R_t^k is the amount of release resulting in power generation at time t, from reservoir k, and $R_t'^k$ represents all other releases.

A number of linear programs are solved for each decision period. The minimum on-peak energy constraint is varied parametrically upward from the contract level for each decision period and for each beginning storage vector (the ending storage vector for the preceding decision period; these storage vectors will progressively increase with the decision period). Thus the total number of linear programs to be solved may be large, but their solution times are very short.

Clearly, the process must terminate with an acceptable ending storage vector, since project contracts must be met over future time periods. Consequently, bounds are placed, both on the upward parametric variation of the required on-peak generation and the terminal storage vector.

The total number of linear programs to be solved is made manageable by use of a dynamic programming routine in which the cumulative on-peak generation is the state variable, the value of the on-peak energy parameter the decision variable, and the objective function is the weighted sum of the storage vector components. That is, the functional equation is given by

$$V_{t+1}(x_{t+1}) = \max_{u_t} \left[\sum_k \Delta S_{t+1}^k(x_{t+1}, u_t) + V_t(x_{t+1} + u_t) \right] \tag{39}$$

where ΔS^k is the change in storage of the kth reservoir (weights of unity assumed), x represents the accumulated energy, and u is the value of the parameterized energy constraint for any particular decision stage. The dynamics for this model is

$$f(x_t, u_t) = x_t + u_t \tag{40}$$

The calculation ends with a number of optional optimal paths. Each represents the generation of energy equal to or better than the contract level in an optimal way, i.e., a best ending storage vector. Of course, the greatest energy generated will be associated with the least desirable storage vector. The reservoir manager will choose one of these operational policies in accordance with his degree of conservatism and with anticipated or forecast streamflows.

Linear programming therefore serves as the medium for evaluating the OCP problem loss function. Through this analysis, Becker and Yeh are able to provide monthly operating rules for the Shasta-Trinity subsystem of the Central Valley Project consisting of 10 decision variables. *Becker et al.* [1976] apply the same technique to an enlarged (22 decision variables) version of the same subsystem. They also report providing daily and hourly rules, in addition to their monthly OCP policy. Further developments are related by *Yeh and Becker* [1979].

We will mention that *Tauxe et al.* [1979*a, b*] have initiated studies into multiobjective dynamic programming for reservoir operations. It is beyond the scope of this study to discuss the motivation or rationale of multiobjective methods, since they lie outside the classical OCP framework. On the other hand, these multiobjective dynamic programming problems often eventually fall into the OCP framework once sufficient structure is placed on the problem so that solutions are uniquely defined. In particular, Tauxe et al. reduce the

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multiobjective problem to one in which a certain objective is minimized while the others are maintained below certain threshold levels. At this point, conventional dynamic programming ideas are applicable.

A summary of the preceding OCP models for reservoirs is given by Table 5.

Stochastic Case

The origins of dynamic programming, inventory theory, and reservoir management are intimately intertwined. *Masse* [1946] is said (by *Arrow et al.* [1958, pp. 12, 13], *Hadley and Whitin* [1963, p. 364], and *Sobel* [1975, p. 720]) to be the first to achieve a satisfactory solution to an inventory problem with nonnegative variables. Masse's study was concerned with reservoir operations, and he employed the functional equation approach which lies at the heart of dynamic programming.

The earliest reservoir operation study in the English language appears to be the work of *Little* [1955], who considered operation of a single reservoir. The second sentence in his study is, 'The scheduling of the use of this stored water makes an unconventional inventory problem.' He later compares the reservoir operations and inventory problems, writing,

> The hydroelectric problem differs from the usual business inventory problem in that the input, not the output, is the random variable. The power demand is considered fairly well-known in advance, but the river flow is not. Furthermore, reservoirs, unlike most warehouses, have the property that the more nearly they are filled, the more valuable is each unit of water, because head is higher.

In Little's model as well as in most of the models for stochastic reservoir operation, the departure from the model for deterministic reservoir operation is that the inflows $\{i_t\}$ in the dynamic equation (32) are here assumed to be observations of a stochastic sequence, whereas previously we had considered them to be fixed, specified numbers. An important controversy in the literature of stochastic reservoir operations, as well as that of river flow modelling, concerns the appropriate statistical assumptions for this inflow sequence. The appropriateness of a stochastic sequence assumption will depend on the interval between decision times. *Little* [1955] took this time lapse to be two weeks and chose the Markov assumption that the conditional densities for the inflows should satisfy

$$p(i_t|i_{t-1}, \cdots, i_1) = p(i_t|i_{t-1}) \tag{41}$$

The Markov assumption was not supported by statistical analysis, but Little mentions that the much more convenient independence assumption, i.e., that

$$p(i_t|i_{t-1}, \cdots, i_1) = p(i_t) \tag{42}$$

was discarded because it is '. . . untenable for river flow. The flow this week is an indicator of next week's flow, since a drainage basin charged with water will maintain river flow for some time.' These remarks are cogent because, as will be seen, many authors we will cite have adopted the independence assumption (42). The independence assumptions leads to much more elegant and easily computed solution strategies. This author [*Yakowitz*, 1973, 1979; *Denny et al.*, 1974; *Denny and Yakowitz*, 1978] has participated in statistical studies testing the Markov assumption on rivers of the southwest. Here changes in flows (especially those due to summer thunderstorms) appear to have fairly short memories, and on a weekly basis the independence assumption may be tenable; but on a daily basis, it is not.

The Markov assumption and the fact that the loss function depends on reservoir storage s_t (as in the previous section) allow *Little* [1955] to define the state to be the two-dimensional quantity

$$x_t = (s_t, i_{t-1}) \tag{43}$$

His functional equation may be expressed in the standard form (8). To do so, as in the deterministic reservoirs case, one defines

$$s_{t+1} = \min \{s_t + i_t - u_t, C\} \tag{44}$$

C being the capacity of the reservoir. With s_{t+1} so determined, it is clear how one may find its probability density, given u_t, $x_t = (s_t, i_{t-1})$, and the Markov transition density $p(i_t|i_{t-1})$.

Little [1955] applies his model to data from the Grand Cooley generation plant on the Columbia River. The time horizon N is taken to be 26, and the interval between decision times is 2 weeks. Stochastic discrete dynamic programming is employed, and the transition matrix for $p(i_t|i_{t-1})$ is inferred from 39 years of historical flows. The highly nonlinear single-stage loss function for the numerical study reflects the amount of water, at a given head, required to generate a given amount of electricity, and the cost of failing to meet a specified demand. The optimal strategy having been computed, its performance is compared to that of the 'rule curves' then in use, over the 39-year historical record. It is found that the relative improvement using the optimal strategy was only about 1%, but even that figure represents a substantial financial savings.

Inventory theory results concerning the structural forms of optimal strategies are very appealing aesthetically and computationally. While he noted connections with inventory theory, *Little*'s [1955] solution was a computational tour-de-force; no strategy properties were asserted, and no inventory theory methodologies were employed. Because of the complications arising from the Markov inflow model, to this date, it is not known whether optimal strategies to his problem have any useful analytic structure.

On the other hand, *Gessford and Karlin* [1958], through adopting independence assumptions for the inflow process, are able to achieve some inventorylike results. Because of the independence assumption, one may take the state x_t to be the reservoir volume. They assume in their model 1 that the single-stage loss is

$$L_t(x_t, u_t) = \alpha \max \{0, D_t - u_t\} \tag{45}$$

where D_t is the known demand at time t. If $D_t - u_t \geq K$, then α, the constant of proportionality, is increased. The losses are discounted. Under these circumstances, by means of an inventory-type functional equation proof, they demonstrate that the optimal strategy S_t, at any time t, has the form portrayed in Figure 6*a*. The parameters R_1 and R_2 depend on stage t, costs, and reservoir capacity.

In their model 2, *Gessford and Karlin* [1958] assumed that the loss function satisfies the following weaker conditions:

1. The costs incurred are determined by a convex function of the amount of water used to generate power during that decision time interval.
2. All power produced can be sold at a profit.
3. All costs are sufficiently differentiable.

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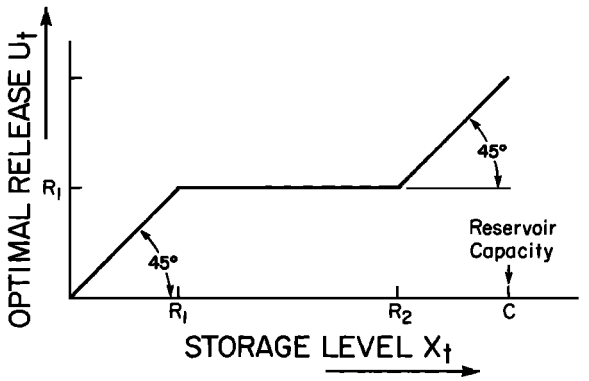


Fig. 6*a*. Optimal strategy structure for model 1.

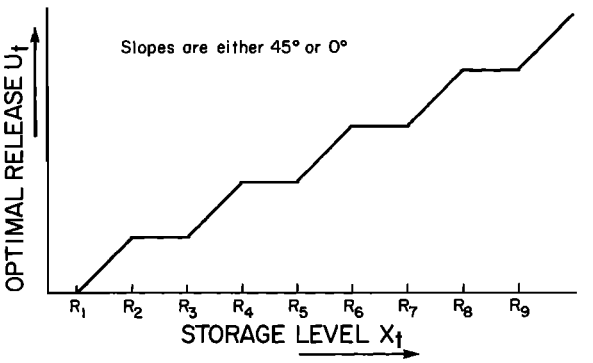


Fig. 7. Optimal strategy structure for Russell's model.

Under these circumstances, *Gessford and Karlin* [1958] have demonstrated that the optimal strategy has the form shown in Figure 6*b*. This result seems of less importance than the strategy for model 1 inasmuch as the shape of the curve above the parameter R_1 is still not specific.

Russell [1972] has extended *Gessford and Karlin*'s [1958] model 1 to include costs related to releases and reservoir free-board passing outside desired intervals. To account for costs due to release u_t being outside some interval $[\bar{u}, \bar{\bar{u}}]$, the authors append to the single-stage loss function $L_t(x_t, u_t)$ the term

$$A \max [(u_t - \bar{u}), 0] + B \max [(\bar{\bar{u}} - u_t), 0] \qquad (46)$$

A similar term involving the reservoir storage x_t is also appended. Otherwise the assumptions of Gessford and Karlin's model 1 are presumed to be in force. Through analysis of the associated dynamic programming functional equation, Russell has proved that at each decision time t the optimal strategy is nondecreasing and piecewise linear, as shown in Figure 7.

It is to be noted that the aim of *Little*'s [1955] dynamic programming application was to find a numerical procedure for obtaining a specific answer to a poorly structured problem, whereas *Gessford and Karlin* [1958] and *Russell* [1972] used the dynamic programming functional equation not to obtain numerical answers, but as an analytic tool to obtain structural results about the character of the optimal strategy. *Bellman* [1957] has stressed the value of dynamic programming as a tool for deriving the structure of optimal policies.

Butcher [1971] adopts the same model (including the Markov inflow process) as *Little* [1955]. He notes that if the terminal decision time is far enough into the future, then (at least for finite state OCP's) one obtains a 'stationary optimal strategy,' i.e., $S_t(x)$ is the same for all t. (Incidentally, such results should be credited to Bellman (e.g., chapter 4 of *Bellman* [1957]), instead of *Howard* [1960], as Butcher does. Howard's contributions are successive approximation schemes for solving certain types of OCP's.) *Butcher* [1971] applies discrete stochastic dynamic programming to find the optimal stationary strategy for operating the Wataheamu Dam on the California-Nevada border. It may be recalled that *Dudley and Burt* [1973] employ stationary strategy analysis for an irrigation problem.

Su and Deininger [1974] apply the model and methodology of *Little* [1955]. Data from Lake Superior served as a basis for their computations, and they obtained a strategy which gave a relative improvement of 15% over the policy used by the Army Corps of Engineers. A feature of *Su and Deininger* [1974] is the use of time dependence of the Markov transition probabilities in order to capture the effects of seasonality.

Gablinger and Loucks [1970] also consider the OCP model for reservoir operation proposed by *Little* [1955]. They note that after states and controls have been discretized, the OCP can be solved by linear programming (or more precisely, the 'simplex method') as well as dynamic programming. One allows that 'randomized strategies' can be used. That is, at each state and time, the control u is chosen by a random mechanism. (See *Lindgren* [1971], for example, for elaboration on randomized strategies.) The usual dynamic programming strategies may be viewed as 'degenerate' cases of randomized strategies in which for each state and time, a certain control is chosen with probability 1. Define

$$\theta_t(x, u) = \text{joint probability of state } x \text{ and control } u \text{ at time } t \qquad (47)$$

These values $\{\theta_t(x, u)\}_{x,u}$ are the variables to be obtained by linear programming so as to minimize the objective function

$$\sum_{t=1}^{N} \sum_{x,u} L_t(x, u)\, \theta_t(x, u) \qquad (48)$$

This objective function is linear in the variable $\theta_t(x, u)$. It turns out that the constraints imposed to assure that $\{\theta_t\}$ really is a probability function, and also that the reservoir mass balance equations hold, are also linear; the main constraints are

$$\theta_t(x, u) \geq 0 \qquad \sum_{x,u} \theta_t(x, u) = 1 \qquad (49)$$

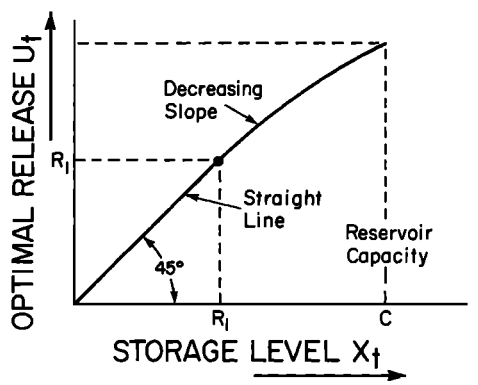


Fig. 6*b*. Optimal strategy structure for model 2.

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and, in terms of the probabilistic state transition function p,

$$\sum_{\bar{u}} \theta_{t+1}(\bar{x}, \bar{u}) = \sum_{x,u} \theta_t(x, u)p(\bar{x}; x, u) \tag{50}$$

It is to be observed that Bellman [e.g., *Bellman and Dreyfus*, 1962, p. 319] had already noted that linear programming is applicable to stochastic OCP's. The question of practical importance is whether linear programming is more efficient than dynamic programming in obtaining optimal strategies. *Gablinger and Loucks* [1970] assert,

> For the simplified example problem presented herein the dynamic programming models defined a steady-state policy in about 1/20 the time required by the linear programming models.

In the opinion of the author, the reason for the linear programming method's relatively poor performance is that the linear programming transcription requires many more solution variables than does dynamic programming. Specifically, the number of Θ_t values to be determined by the linear programming solution equals the product of the number of discretized points in state and policy spaces, multiplied by the number of decision times, whereas the number of variables (namely, controls for each state and time) to be solved for in the discrete dynamic programming formulation equals only the product of the number of state values multiplied by the number of decision times. If there are 10 decision times and, as in the *Little* [1955] model, a bivariate state space and if everything is discretized into 10 levels, the number of solution variables to be obtained by dynamic programming is 10^3, whereas the number of variables to be found by linear programming is 10^4.

Schweig and Cole [1968] adapt the *Little* [1955] model to a two-reservoir problem which, structurally, is essentially the same as that of *Buras*' [1972] conjunctive reservoir-aquifer model discussed in the section 'miscellaneous application.' In particular, Figure 3 describes the interrelations of the variables in the work of *Schweig and Cole* [1968]. Through discrete stochastic dynamic programming, Schweig and Cole compute an optimal strategy for a problem based on data from Lake Vyrnwy, Wales. A point to be noted is that despite very coarse discretization (for example, the state coordinates for past inflows were discretized into only two levels), the authors mention severe computational difficulties. To our knowledge, two-reservoir systems are the largest to be reported solved by stochastic dynamic programming, whereas we have noted that deterministic reservoir systems of up to 10 reservoirs have been solved. This observation points to the motivation for making the deterministic assumption and underscores the need for research ideas for overcoming the computational burden of the stochastic case.

The final reservoir modelling issue to be addressed in this section is incorporation of 'chance constraints.' *Askew* [1974*a*, *b*, 1975] introduced this line of study to the realm of nonlinear reservoir operations models. But as Askew noted, there are several papers [e.g., *Eisel*, 1972; *Joeres et al.*, 1971; *Revelle et al.*, 1969] discussing chance constraints in connection with linear reservoir models. Askew and other authors we will mention assumed the random inflows $\{i_t\}$ to be independent (and thus reservoir content alone was the state). But otherwise he adopted *Little*'s [1955] basic model of dynamics and loss. In the 'chance-constrained' reservoir operations model, one seeks to minimize

$$J(\mathbf{S}) = \sum_{t=1}^{N} E[L_t(s_t, s_{t+1}, S_t(x_t))] \tag{51}$$

subject to the 'target release' constraint that $u_t \geq D_t$, whenever possible and the 'chance constraint' that

$$P[s_t + i_t \geq D_t, 1 \leq t < N] > 1 - \beta \tag{52}$$

where β is a specified constant. Askew proposed a strategy constructed by functional equation-type recursions. While the strategy so constructed does satisfy the chance constraint, it does not necessarily achieve the minimizing value of J among all such constrained strategies.

Askew also proposed variations on his chance constraint, such as that the expected number of failures over the operations horizon be bounded. It does not seem possible to fit such chance-constrained reservoir problems exactly into the OCP format described in section 2 of this survey. On the other hand, dynamic programming techniques can play a vital step in the solution algorithm, as we will see.

Sniedovich and Davis [1975] noted a number of difficulties with *Askew*'s [1974*a*, *b*] developments. They reported that the chance-constrained reservoir operation problem could be put into the stochastic OCP framework by augmenting the state variable x_t so that

$$x_t = (s_t, y_t) \tag{53}$$

where y_t is defined as the 'probability of no failure up to the year under consideration' and s_t is reservoir storage at time t.

Sniedovich and Davis [1975] did not demonstrate that x_t, so defined, is a proper state variable in the sense that the distribution of x_{t+1} depends on the history $(x_t, u_t, x_{t-1}, \cdots, x_1)$ only through (x_t, u_t). By way of correspondence with Dr. Sniedovich, we have learned that the interpretation of 'chance constraint' he had in mind was not our interpretation (52) of *Askew* [1974*b*], but rather that

$$P[s_t + i_t \geq D_t, 1 \leq t \leq N | s_1, s_2, \cdots, s_{N+1}] > 1 - \beta \tag{54}$$

With this interpretation, Sniedovich can prove that x_t defined by (53) is a proper state variable. On the other hand, the constraint (54) does not seem intuitively satisfying. For example, if the inflows are continuously distributed, then it is simple to see that failure occurs if and only if at least one of the s_t's in the conditioning is 0. Thus the only possible values for the probability are 0 and 1.

The condition (52) that, prior to beginning the decision process, the probability of failure not exceed β seems more natural. For this criterion, using techniques such as in the work of *Sniedovich* [1981], Sniedovich has established a deterministic OCP model in which one of the state coordinates takes values in function space (instead of the real numbers, as we have assumed). Such a model is not at all suited to computations, of course, but parameterizations of function space may make this approach feasible.

Rossman [1977] applied Lagrangian duality theory to the reservoir operations problem in which the expected number of failures is constrained to not exceed a given constant β. Assume that state space has been discretized so that a

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TABLE 6. Applications to Stochastic Reservoir Operations

Author(s)	Subject	State	Control	Components of Single-Stage Loss	Basis for Dynamics
Little [1955] *Butcher* [1971] *Su and Deininger* [1974]	Operation of a single reservoir	Storage and prior inflow	Release	Satisfaction of power demand	Water mass balance, Markovian inflows
Gessford and Karlin [1958]	Operation of a single reservoir	Storage	Release	Linear cost of unmet power demand	Mass balance, independent inflows
Russell [1972]	Operation of a single reservoir	Storage	Release	Linear cost of unmet demand and release or storage falling outside specified bands	Mass balance, independent inflows
Schweig and Cole [1968]	Operation of two reservoirs subject to chance constraints	Storage and prior inflow	Release	Unspecified functions of release and storage	Mass balance, independent inflows
Askew [1974*a*, *b*, 1975] *Rossman* [1977]	Operation of a single reservoir subject to probabilistic constraints	Storage	Release	Unspecified functions of release and storage	Mass balance, independent inflows
Sniedovich [1979]	Operation of a single reservoir subject to probabilistic constraints	Storage and counts of reservoir emptiness	Release	Unspecified functions of release and storage	Mass balance, independent inflows

strategy S can be represented by a finite dimensional vector. Then the Lagrangian for the chance-constrained problem can be written as

$$\mathcal{L}(\mathbf{S}, \lambda) = \sum_{t=1}^{N} E[L_t(x_t, x_{t+1}, S_t(x_t)) + \lambda(r_t(x_t) - \beta)] \qquad (55)$$

where $r_t(x_t)$ is 1 or 0, according to whether x_t is equal to or greater than 0. Define

$$H(\lambda) = \min_{\mathbf{S}} \mathcal{L}(\mathbf{S}, \lambda) \qquad (56)$$

From Lagrangian duality theory references which Rossman cited, it is known that under certain convexity conditions, if λ^* is the maximizer of $H(\lambda)$, and $\mathbf{S}^*$ is the strategy minimizing $\mathcal{L}(\lambda^*, \mathbf{S})$, then $\mathbf{S}^*$ is the solution to the constrained optimization problem. If the requisite convexity conditions are not satisfied, there are procedures available to 'close the duality gap' and approximate the solution. Rossman outlined and cited some procedures. The idea of using Lagrangians for the dynamic programming solution of chance-constrained OCP's had already been explored by *White* [1974] and others.

The problem of finding the minimizing strategy, for any fixed λ, is a stochastic OCP problem which differs from the unconstrained model only in that the single-stage loss function is now given by

$$\tilde{L}_t(x_t, x_{t+1}, u_t) = L_t(x_t, x_{t+1}, u_t) + \lambda(r_t(x_t) - \beta) \qquad (57)$$

Rossman calls upon stochastic dynamic programming to solve for the minimizing strategy at each iteration, as he sequentially searches for the value λ maximizing $H(\lambda)$. In summary, while the chance-constrained reservoir operations problem has not yet itself been modelled as a stochastic OCP (with real state variables), its solution can be found by searching over a family of OCP solutions.

Rossman computed optimal policies to chance-constrained discounted reservoir problems investigated by *Askew* [1974*b*]. The performance of strategies he constructed is better than those using Askew's heuristic algorithm. Also, from Rossman's numerical results, one sees that with chance-constrained programming, in the presence of 'duality gaps,' advantage can be gained by employing randomized strategies, an advantage previously noted by *Sobel* [1975]. *Askew* [1978] remarked that the chance-constrained problem Rossman solved was not exactly the one he had in mind.

Sniedovich [1979, 1980] has shown that reservoir control under probabilistic and under variance constraints also are amenable to the algorithm Rossman has proposed. In fact, as *Sniedovich* [1979] observed, probabilistic constraints can be viewed as expectation constraints. Introduction of a state variable to count the number of failures is, however, necessary.

Table 6 summarizes the stochastic OCP reservoir models discussed above.

5. COMPUTATIONAL ASPECTS

Deterministic Case

As was discussed in section 2 of this survey, the exponential growth in memory and CPU time requirements with increase in dimension of the state vector (i.e., the 'curse of dimensionality') is the greatest single hindrance to dynamic programming solution of large-scale optimal control problems. The appropriateness of a given numerical procedure depends strongly on whether the deterministic OCP to be solved has a continuous or discrete state space. Problems of project sequencing and aqueduct design (assuming only pipes with certain fixed diameters are available) would seem to fall naturally into the latter category, while irrigation, reservoir, and water quality control typically are continuous state problems. To keep the scope of this survey within bounds and because of the author's predilections, our discussions will be restricted to methods for the continuous state problems. The reader is referred to certain contributions in the work of *Puterman* [1978] (especially the survey by Morin) for an introduction to the literature of modern discrete-state-space techniques such as fathoming and

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reaching. Our discussions will also overlook many pragmatic schemes which restrict the class of policies for purposes of computational convenience, even though the restricted set may not contain the optimal policy. The 'myopic' reservoir operations algorithm by *Croley* [1974] is an example of such an approach: for some fixed integer k, the control at stage t is chosen to be optimal if only k stages remain, regardless of the actual number $N - t$ of remaining stages.

When authors discuss the continuous state case in a numerical context, they really mean more: the single-stage loss function and the dynamics must be differentiable with respect to state and control. Otherwise convergence cannot be assured. What this means to the practitioner is that corners and steps must be approximated by smooth humps and rises. A tradeoff between modelling perfection and computational tractability confronts engineers, economists, hydrologists, and all others engaged in solution of messy quantitative problems. The modeller who refuses to compromise will not be able to find optimal policies, and the theorist with his computational example will obtain the optimal policy to an unrealistic problem.

The three most highly regarded methods advanced in the literature to lessen the computational burden are (1) differential dynamic programming (DIFF DP), (2) discrete differential dynamic programming (DDDP), and (3) state incremental dynamic programming (IDP).

These methods will be outlined and discussed in order of this author's increasing enthusiasm, DDDP being the least interesting, and DIFF DP being, in his opinion, the most promising. All these schemes are 'successive approximation' algorithms. That is, an initial estimate $\mathbf{u}^{(1)}$ of the optimal policy must be supplied by the user. The successive approximation algorithm describes how to construct from $\mathbf{u}^{(1)}$ a new policy, $\mathbf{u}^{(2)}$, which, in turn, serves as an 'input' policy from which $\mathbf{u}^{(3)}$ is constructed. Thus policies $\mathbf{u}^{(2)}$, $\mathbf{u}^{(3)}$, $\cdots$, are successively constructed according to the algorithm until some convergence criterion is satisfied.

A drawback of even the best successive approximation methods is that the policy sequence $\{\mathbf{u}^{(n)}\}$ can converge to local optima, or may not converge at all, unless the optimal control problem itself satisfies somewhat stringent assumptions.

For notational convenience, we will let the 'nominal policy' $\mathbf{u}^{(1)}$ supplied by the user be denoted by $\bar{\mathbf{u}}$. To describe a successive approximation method, it is sufficient to describe the construction of the successor policy $\mathbf{u}^{(2)}$. We shall let $\mathbf{u}'$ denote this successor. Let $\bar{\mathbf{x}} = \{\bar{x}_t\}$ denote the trajectory determined by the nominal policy $\bar{\mathbf{u}}$ through the dynamics (1). Similarly, $\mathbf{x}'$ is the trajectory determined by the successor policy $\mathbf{u}'$.

The DDDP and IDP methods are most readily described in terms of discrete dynamic programming (DDP). At this point the reader may wish to recall the discussion of DDP given in section 2. Prototypical DDDP is most simply described as DDP with the added constraint that for each t and for some specified $\varepsilon > 0$,

$$\|\bar{x}_t - x_t'\| < \varepsilon \tag{58}$$

The norm in (58) is the uniform norm; that is, if $a = (a(1), \cdots, a(n))$, then $\|a\| \max_i |a(i)|$. The constraint (58) has the capability of vastly reducing the number of discretized control values that must be tested at each DDDP iteration, over the requirements of DDP (but, of course, DDP is only applied once, whereas DDDP is a successive approximation procedure). *Heidari et al.* [1971] describe the constraint (58) by saying that the successor trajectory must lie in a 'corridor' of width ε centered about the nominal trajectory. The DDDP procedure was first described by *Hall et al.* [1969*a*, *b*] who, unfortunately, referred to it as incremental dynamic programming, an appellation which, as we will presently see, has come to denote a different scheme. The designation 'DDDP' is due to *Heidari et al.* [1971]. See *Chow et al.* [1975] and *Nopmongcol and Askew* [1976] for applications and further developments of this technique.

To get some feeling for the properties of DDDP, let us consider the simple OCP case in which state space is the unit n cube (i.e., each coordinate has to lie between 0 and 1 and the dimension of the state vector is n). Suppose further that the state coordinates are discretized by a uniform mesh of size 0.01. Let $\varepsilon = 0.015$ and suppose the dynamics, as a function of u, $f_t(x, u)$ is one-to-one. Then for each decision stage t and state x, under DDDP there will be 3^n control values that must be searched over to minimize $V_t(x^{(i)})$ in (6), whereas DDP itself will require that approximately $(101)^n$ control values be searched.

Thus it is clear that a single DDDP iteration will typically be much less expensive than a DDP solution. Note, however, that the DDDP effort still grows exponentially with state dimension n and, so like DDP, suffers the 'curse of dimensionality.' As we will see, IDP and DIFF DP do sidestep the 'curse.'

The (state) incremental dynamic programming (IDP) method, due to *Larson* [1968, chapter 12] and *Korsak and Larson* [1970], does have the potential for overcoming the 'curse of dimensionality.' In the form described in the above citations, the IDP method requires that for each x and t, the dynamics $f_t(x, u)$ be invertible in u. That is, there must exist a function $h_t(x, y)$ such that for every pair of states x_t and x_{t+1},

$$h_t(x_t, x_{t+1}) = u_t \tag{59}$$

where u_t is the (necessarily unique) control such that $x_{t+1} = f_t(x_t, u_t)$. For purposes of description of this method, let $x(i)$ denote the ith coordinate of the state vector x. With this notation, one can describe the IDP method as being an application of DDP to the OCP, with the added constraint that for some specified coordinate i,

$$x_t'(j) = \bar{x}_t(j) \tag{60}$$

for all $j \neq i$ and all $t = 2, \cdots, N + 1$. That is, in a given IDP approximation, in passing from $\bar{\mathbf{u}}$ to $\mathbf{u}'$, the only state coordinate that is allowed to change is the ith coordinate. One may note that the constraint (60) can be put into the canonical form (3), and so an IDP iteration does constitute an OCP. The important feature is that the constraint (60) reduces the dimensionality of the state space of the OCP problem from n to 1. In successive IDP iterations the value of the fixed state variable i changes. *Korsak and Larson* [1970] recommend that it be increased by one until it reaches n and then reduced to 1 again, etc.

To emphasize that an IDP iteration is essentially a DP solution of an OCP with a univariate state space, observe that in view of (60) any successor state x_t at stage t must have the representation

$$x_t = x_t(\alpha_t) = x_t + \alpha_t e_i$$

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e_i being the unit vector with 1 in the ith coordinate and 0's elsewhere, and α_t being a real scalar. In this notation, if $\hat{V}_t(\alpha_t) \triangleq V_t[x_t(\alpha_t)]$, the basic functional equation (4) can be expressed as

$$\hat{V}_t(\alpha_t) = \min_{\alpha_{t+1}} \{L_t[x_t(\alpha_t), h_t(x_t(\alpha_t), x_{t+1}(\alpha_{t+1}))] + \hat{V}_{t+1}(\alpha_{t+1})\} \tag{61}$$

Thus the minimization is accomplished by a line search and the optimal return function $\hat{V}_t$ is indexed by a single real variable. From these considerations, one may demonstrate that the computational burden and essential memory requirements of an IDP iteration do not depend on the state dimension: IDP does overcome the 'curse of dimensionality.'

In most of his many works on reservoir operations, Yeh has mentioned using IDP in at least portions of his algorithms. *Trott and Yeh* [1973] describe the details of the IDP method and its application to reservoir operations. The reservoir operations problem would seem especially well suited to IDP application because, between extreme states, $f_t(x, u)$ is an invertible linear function of u. One can describe an IDP iteration to the multireservoir problem simply as, 'optimize the release of the ith reservoir while holding the contents of the remaining reservoirs constant over the decision horizon.'

With successive approximation methods such as DDDP, IDP, and DIFF DP, crucial questions arise that are absent in the basic DP structure. There are two particularly important questions about a successive approximation method:

1. Do the successive policy iterates $\{\mathbf{u}^{(n)}\}$ converge, and if so, to what?
2. Assuming convergence, what is the convergence rate?

Ignoring the effects of discretization of state space, one can prove that under various smoothness assumptions, DDDP, DIFF DP, and IDP policy iterates do converge, with decreasing discretization grid size to 'stationary policies,' i.e., policies for which the gradient of the objective function $J(\mathbf{u})$ is zero, or nonnegative in all feasible directions. This is the strongest sort of statement one can make concerning any successive approximation scheme, without imposing stringent conditions such as convexity on the OCP.

Yakowitz [1981] has demonstrated that (still neglecting the effect of discretization) the IDP method coincides with a technique for nonlinear equations known as the 'generalized block Gauss-Seidel' method. This insight implies, in view of what is known about Gauss-Seidel methods, that the convergence rate of the IDP algorithm must be linear. That is, for some fixed number C less than one and all n, and $\mathbf{u}^*$ the limiting stationary policy,

$$\|\mathbf{u}^{(n+1)} - \mathbf{u}^*\| < C \|\mathbf{u}^{(n)} - \mathbf{u}^*\| \tag{62}$$

The generalized Gauss-Seidel method, in particular, and linearly convergent methods, in general, are not widely esteemed for computational efficiency.

DIFF DP (like IDP) overcomes the 'curse of dimensionality' and furthermore achieves quadratic covergence. That is, in the terminology related to (62), for some fixed C and all n, under DIFF DP,

$$\|\mathbf{u}^{(n+1)} - \mathbf{u}^*\| \leq C \|\mathbf{u}^{(n)} - \mathbf{u}^*\|^2 \tag{63}$$

Quadratic convergence is the fastest achieved by any of the popular successive approximation methods for nonlinear optimization. The DIFF DP method for unconstrained OCP's was proposed independently by *Jacobson and Mayne* [1970, chapter 4 and references therein] and *Dyer and McReynolds* [1970, pp. 68–79]. *Murray and Yakowitz* [1979] have extended the method to OCP's with linear constraints. This is sufficient for applications to the deterministic reservoir model.

The implementation of DIFF DP is a little more involved than that of DDDP or IDP. For simplicity of notation, we assume that the dimension of state and control spaces is one. However, it should be fairly clear how to generalize the construction to follow to higher dimension; the details of the extension can be obtained from *Jacobson and Mayne* [1970], for example. We will also begin by assuming that there are no constraints and then later point out how constraints may be accounted for. Finally, it will be supposed that the dynamics and single-stage loss functions are all sufficiently differentiable that all terms to follow exist. We continue to let $\bar{\mathbf{u}}$ and $\bar{\mathbf{x}}$ denote the nominal policy and the associated trajectory. The notation

$$\delta x_i = x_i - \bar{x}_i$$
$$\delta u_i = u_i - \bar{u}_i$$

will be useful. We begin by defining $Q_N(x_N, u_N)$ to be the quadratic part of the Taylor's series expansion of $L_N(x_N, u_N)$, the expansion being about the nominal values $\bar{x}_N$ and $\bar{u}_N$. The coefficients of the expansion are labeled as

$$Q_N(x_N, u_N) = \tfrac{1}{2}D_N\delta x_N^2 + E_N\delta x_N\delta u_N + \tfrac{1}{2}F_N\delta u_N^2 + G_N\delta x_N + H_N\delta u_N \tag{64}$$

The constant term is omitted from (64) and quadratics to follow because it has no effect on the choice of optimal policy. The condition of vanishing derivative implies that u_N is the optimal control under single-stage loss function Q_N at state x_N if and only if

$$u_N(x_N) = \bar{u}_N + \delta u_N \tag{65}$$

where

$$\delta u_N = \alpha_N + \beta_N\delta x_N \tag{66}$$

with

$$\alpha_N = -F_N^{-1}H_N \qquad \beta_N = -F_N^{-1}E_N \tag{67}$$

The approximate optimal value function $\hat{V}_N(x_N)$ is determined by

$$\hat{V}_N(x_N) \triangleq Q_N[x_N, u_N(x_N)] = (1/2)A_N(\delta x_n^2) + B_N\delta x_N \tag{68}$$

If we evaluate the coefficients of the quadratic (68), we find that

$$A_N = D_N - E_NF_N^{-1}E_N$$
$$B_N = G_N - E_NF_N^{-1}H_N \tag{69}$$

The general step at decision stage t follows a very similar construct. Assume that the approximate optimal return quadratic $\hat{V}_{t+1}(x_{t+1}) = (1/2)A_{t+1}(\delta x_{t+1})^2 + B_{t+1}\delta x_t$ has been obtained already. Define $Q_t(x, u)$ to be the quadratic part of the Taylor's series expansion of $L_t(x_t, u_t) + \hat{V}_{t+1}[f_t(x_t, u_t)]$

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about $(\bar{x}_t, \bar{u}_t)$. Note the motivation for this approximation in light of (4). Then one may represent $Q_t(x_t, u_t)$ as

$$Q_t(x_t, u_t) = \tfrac{1}{2}D_t\delta x_t^2 + E_t\delta x_t\delta u_t + \tfrac{1}{2}F_t\delta u_t^2 + G_t\delta x_t + H_t\delta u_t \quad (70)$$

where

$$D_t = \frac{\partial^2 Q_t}{\partial x_t^2} = \frac{\partial^2 L_t}{Qx_t^2} + A_{t+1}\left(\frac{\partial f_t}{\partial x_t}\right)^2 + B_{t+1}\frac{\partial^2 f_t}{\partial x_t^2}$$

$$E_t = \frac{\partial^2 Q_i}{\partial x_t \partial u_t} = \frac{\partial^2 L_t}{\partial x_t \partial u_t} + A_{t+1}\frac{\partial f_t}{\partial x_t}\frac{\partial f_t}{\partial u_t} + B_{t+1}\frac{\partial^2 f_t}{\partial x_t \partial u_t}$$

$$F_t = \frac{\partial^2 Q_t}{\partial u_t^2} = \frac{\partial^2 L_t}{\partial u_t^2} + A_{t+1}\left(\frac{\partial f_t}{\partial u_t}\right)^2 + B_{t+1}\frac{\partial^2 f_t}{\partial x_t \partial u_t} \quad (71)$$

$$G_t = \frac{\partial Q_t}{\partial x_t} = \frac{\partial L_t}{\partial x_t} + B_{t+1}\frac{\partial f_t}{\partial x_t}$$

$$H_t = \frac{\partial Q_t}{\partial u_t} = \frac{\partial L_t}{\partial u_t} + B_{t+1}\frac{\partial f_t}{\partial u_t}$$

are evaluated at $(\bar{x}_t, \bar{u}_t)$.

With the coefficients Q_t thus determined, one now proceeds as in the case $t = N$ to find the coefficients α_t and β_t, of the optimal control $u_t(x_t)$ and A_t and B_t of the approximate optimal return function $\hat{V}_t(x_t)$. As in the $t = N$ case, the condition of vanishing derivative of Q_t with respect to u_t implies that

$$u_t(x_t) = \bar{u}_t + \alpha_t + \beta_t\delta x_t \quad (72)$$

where

$$\alpha_t = -F_t^{-1}H_t \qquad \beta_t = -F_t^{-1}E_t \quad (73)$$

and the coefficients A_t and B_t of the approximate optimal return function $\hat{V}_t(x_t) = \frac{1}{2}A_t(\delta x_t^2) + B_t\delta x_t$ are given by (69), with subscript t replacing N. The numbers α_t and β_t must be stored for every stage t, but the coefficients $A_{t+1}, \cdots, H_{t+1}$ can be erased once the calculations for the coefficients for decision stage t are completed.

When the coefficients α_t and β_t for the approximate optimal strategy $u_t(x_t)$ have been computed for all decision stages, analogously to (5), the successor policy is computed according to

$$u_t' = \bar{u}_t + (\alpha_t + \beta_t\delta x_t') \qquad x_1' = x_1$$
$$x_{t+1}' = f_t(x_t', u_t') \quad (74)$$

The policy $\mathbf{u}' = \{u_t'\}$ is the DIFF DP successor policy.

The DIFF DP method was deemed worthy of its extensive description here because for theoretical reasons (especially the quadratic convergence property (63)) and in view of computational experience, it seems significantly superior to DDDP, IDP, and other computational schemes to come to our attention.

By way of illustration of its computational potential, we mention that *Yakowitz and Rutherford* [1981] have solved OCP's with as many as 40 state variables. Also, *Murray and Yakowitz* [1981] have shown that if certain standard nonlinear programming test problems are reformulated as OCP's, one can obtain the solution with less computational effort through DIFF DP than authors have achieved on the original problems using the very best available nonlinear programming methods.

The extension of DIFF DP to OCP's with linear constraints was accomplished by *Murray and Yakowitz* [1979]. The central construct of the procedure involves replacing the unconstrained minimization of the quadratics $Q_t(x, u)$ with the quadratic programming problem of minimizing the Q_t's subject to the linear constraints. The details of this extension are given in the above-cited paper, and a numerical application to a deterministic 10-reservoir operation problem is described. The performance of DIFF DP on four-reservoir problems is also reported in that paper to be far superior to that obtained by *Heidari et al.* [1971] using DDDP and by *Larson* [1968] using IDP. *Yakowitz* [1982] has supplied a convergence proof for the constrained DIFF DP algorithm.

It should be observed that some large reservoir operations problems have been solved as nonlinear programming problems (rather than OCP's) by non-dynamic-programming methods. *Lasdon and Waren* [1980] include some mention of these works, and *Rosenthal* [1980] gives a large number of citations of both OCP and nonlinear programming assaults on the deterministic reservoir operations problem. The most impressive such nonlinear programming computation was reported by *Gagnon et al.* [1974] and is described with more peripheral detail in the book by *Jacoby and Kowalik* [1980]. The objective function does not fit into the OCP format (2). Unfortunately, not all the details are specified, so one cannot reconstruct their calculation from the publications cited. But it seems fair to say that these computations are the largest, in terms of number of variables yet reported, and that this computational technique should be regarded as a competitive approach to the reservoir operations problem. The method can be roughly described as a projected conjugate gradient technique. The largest of the computations included 38 reservoirs and 49 decision stages. Its solution required an hour of CDC 6400 computer time.

In closing, we note that there is a discrete time version of Pontryagin's maximum principle approach to OCP's. The 'discrete maximum principle,' as this version is known, involves converting the OCP to a two-point boundary value difference equation. This approach [e.g., *Tamura*, 1979; *Halkin*, 1966] has not been popular, but it should not be forgotten, for advances in numerical analysis may yet make it computationally competitive.

Stochastic Case

The situation with respect to stochastic dynamic programming is that there are, as yet, no widely applicable computational devices other than discrete dynamic programming (DDP). Because of the 'curse of dimensionality,' as we have often remarked, DDP is not adequate for solving many water resource problems of interest. The largest numerical stochastic dynamic programming solutions within or outside the water resources literature to come to our attention are for problems having at most two or three state variables. Even then, the authors reporting their findings often remark on the ferocity of the computational burdens. For example, in relating their solution of a two stochastic reservoir operation, *Schweig and Cole* [1968, p. 497] have written, 'The greatest hurdle is that of computer running time, which

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mitigates against too fine a subdivision of our basic starting points, the state vectors.'

The numerical studies of stochastic OCP's surveyed in section 3 all employed some variant of DDP. It is clear that one is motivated to make a coarse discretization of the state space in order to keep the size to a manageable number. However, *Doran* [1975] has found that Markov chains describing reservoir storages give incorrect and even nonsensical answers when the state space (i.e., the interval from [0, C], C being the reservoir capacity) is discretized too coarsely, or when the discretization does not intelligently account for the probability density describing the storages. The reader may note that reservoir storages are a Markov process, even though the inflows may be independent. Thus Doran's observations have universal applicability to reservoir operation models.

Klemeš [1977] noted that faulty Markov models necessarily entail suboptimal DDP solutions to the reservoir operations problems. He showed by example that for a given number of levels of discretization, as the storage capacity becomes large, the computed loss becomes seriously in error. He then derived formulas for the minimum number of discretization levels so that for certain reservoir operation problems, satisfactory approximation of the optimal strategy can be assured. This study by *Klemeš* [1977] should be regarded as a significant pilot study, in the spirit of numerical analysis practice, for the important and overlooked task of error analysis for DDP. One should be wary of proposed solutions until the deleterious effects of discretization have been somehow bounded and found acceptable.

The group of papers which we now discuss are direct progeny of *Gessford and Karlin* [1958], a study described in section 4. Arunkumar has continued the line of attack instituted in model 1 of *Gessford and Karlin* [1958] and by *Russell* [1972], in showing that piecewise linear strategies such as shown in Figure 6 are optimal over a wider range of reservoir operation problems. Such an insight, as noted earlier, allows one to replace the burdensome DDP procedure with that of finding the positions of the corners of the piecewise linear strategy functions, which usually is a much simpler task. In all of Arunkumar's studies, it is presumed that the inflows are independent, identically distributed discretized random variables, and that the single-stage loss functions $L_t(x, u)$ are the same for all t and are piecewise linear and convex. Specifically, for some numbers s, k_1, and k_2, with $k_1 < k_2 < 0$,

$$\begin{aligned} L_t(x, u) &= k_1 u & 0 \le u \le s \\ L_t(x, u) &= k_1 s + k_2(u - s) & s \le u \le C \end{aligned} \tag{75}$$

where x is the reservoir storage, u is the release, and C is the reservoir capacity.

Arunkumar [1975] gives a recursive procedure for determining the strategy corners for a single reservoir so that optimal strategies for infinite horizon operation problems can be approximated. *Arunkumar* [1977] extends the previous work to account for seasonality, and *Arunkumar and Chon* [1978] derive piecewise linear optimal control strategies for a system of two reservoirs, each with inflows, in which the release from one of the reservoirs adds to the inflow of the other.

Sobel [1975] finds highly structured optimal strategies for multireservoir operation problems. The most interesting result is closely allied to developments in model 2 of *Gessford and Karlin* [1958]. Specifically, Sobel assumes that the inflows for a multireservoir system are independent and identically distributed. He introduces a change of variable in order to make the reservoir operations problem closely conform to the inventory problem with 'lost sales.' If x is the vector of reservoir storages, u is the vector of releases, and C is the vector of reservoir capacities, then $v \triangleq C - x$ and $w \triangleq v + u$; these vectors can be viewed as the freeboards before and after releases, respectively. Sobel proves that if, after the above transformation of variables, the single-stage loss function $L_t(v, w)$, for each t, has nonpositive cross partials, i.e., $\partial^2/\partial v^i \partial w^j L_t(v, w) \le 0$, then the optimal release strategy $S_t^*(x)$ satisfies the condition that its Jacobian matrix $\partial/\partial x S_t(x)$ has all coordinates between zero and one. This structural result, of course, can be used to reduce greatly the search involved in implementing DDP, as Sobel demonstrates by example.

There are two further computational stochastic dynamic programming approaches in the hydrology literature which seem worthy of mention. *Gal* [1979] proposes a technique reminiscent of the collocation method for approximate solution of differential equations. Specifically, let $\mathcal{V} = \{v(x, a)\}$ denote a family of real functions defined on state space and indexed by an M dimensional parameter $a = (a^1, \cdots, a^M)$, and let X be a finite subset of state space. Under the approach *Gal* [1979] considers, one picks a_N to be the parameter vector which minimizes $\Sigma_{x \in X}[V_N(x) - v(x, a)]^2$, where N is the final decision stage and, as in (8), $V_N(x)$ is the optimal return function, i.e., $V_N(x) = \min_u \int L_N(x, y, u) P_N(y; x, u) dy$, or a numerical approximation thereof. Thus, in essence, $v(x, a_N)$ is a best approximation of the return function for stage N. The process continues inductively to earlier decision stages. Assume that the parameter a_{t+1} for approximating the return function at stage $t + 1$ has already been obtained. One chooses a_t to minimize

$$\Sigma_{x \in X}[v(x, a) - \hat{V}_t(x)]^2 \tag{76}$$

where for any state x, $\hat{V}_t(x) = \min_u Q_{t+1}(x, u)$, with

$$Q_t(x, u) = \int [L_t(x, y, u) + v(y, a_{t+1})] p_t(y; x, u) dy$$

The parameter vectors $a_1, \cdots, a_N$ of the approximate optimal return functions having been obtained, for any decision stage t and state x, the associated strategy is to choose the control u which minimizes $Q_t(x, u)$. Such approaches had been considered by *Bellman and Dreyfus* [1962, chapter 12] and, like DIFF DP for deterministic OCP's, are promising because they do not involve discretizing the state space and are thus potentially a way to sidestep the 'curse of dimensionality.' On the other hand, it is clear that *Gal*'s [1979] scheme is effective only to the extent that $v(x, a_t)$ does accurately approximate the optimal return function for stage t over substantial portions of the state space. (By contrast, the reason DIFF DP works is that the approximation need be good only in a small neighborhood of the nominal trajectory.) In order to achieve a class rich enough to approximate wide varieties of optimal return functions, the number of parameter coordinates M may have to grow very fast with dimension n of state space, and thus the task of finding the minimizing parameter can become expensive. For example, if one chooses $\mathcal{V}$ to be polynomials up to degree 3, the number of parameters grow as n^3 and for

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$n = 4$ one has 20 parameters already; this entails a formidable minimization problem.

Mawer and Thorn [1974] have considered a heuristic variation of *Howard*'s [1960] policy iteration method, which has also been considered by several other authors [*Bogle and O'Sullivan,* 1979*b*; *Buras,* 1972; *Butcher,* 1971; *Dudley and Burt,* 1973; *Dudley et al.,* 1976] surveyed earlier. For details of the method, we refer the reader to *Howard* [1960] or *Derman* [1970], or to *Bhat*'s [1972, chapter 10] concise summary of the method and citations of related studies. The policy iteration method is intended for finite state stationary Markov decision problems; thus the single-stage loss function L_t must be the same for each stage t, and state and control spaces must be discretized. The policy iteration method is a recursive technique that locates the optimal stationary strategy of discretized OCP's. In fact, the policy iteration technique obtains the exact stationary solution for infinite horizon discounted OCP's in finitely many iterations, and there is a test for optimality so that the user can tell when to stop. The computational drawback to the policy iteration method is that at each recursion, one must solve a system of M linear equations, M being the number of states. Current computer technology limits M to be smaller than 1000; thus if the dimension n of the state space is 3, one is able to discretize each coordinate into at most 10 levels. Consequently, it does not appear to us that the policy iteration approach, in its prototypical form, holds great promise for solution of large-scale stochastic OCP's. *Howard* [1978] himself has recently made the sobering statement that

> Recently, a European mathematician who had spent many years on the theoretical study of Markov decision processes visited me and inquired about the range of application. I replied that I knew of very few practical applications and that I had found only one in the course of my career that I considered really successful.

Two speculative differential equation approaches to stochastic OCP's have come to our attention. The first of these has been called the 'stochastic maximum principle'; the reader is referred to *Haussmann* [1981] for discussion and references. The stochastic maximum principle, like the maximum principle itself, has its roots in the Lagrange multiplier technique. Another technique for exploiting Lagrange multipliers for stochastic OCP's has been given by Wets [e.g., *Rockafellar and Wets,* 1978].

The second approach, whose ideas are now outlined, might be designated as 'controlled diffusion.' For a theoretically inclined reader, these ideas are the most interesting of the approaches described in this section, but they do not provide an easy path to mastery of the computational problem. The groundwork to this methodology lies in Markov analysis studies on reservoir problems by *Anis and Lloyd* [1972], *Cohen and Rubinovitch* [1977], *Gani* [1957], *Kendall* [1957], *Lloyd* [1967, 1978], *Moran* [1954, 1959], and *Srinivasan* [1974], among others. The above studies seek to answer probabilistic rather than optimization questions. That is, the investigators were interested in finding the distribution of time until a dam becomes empty and the stationary distribution of storage levels, given some fixed stationary control strategy. *Bather* [1968] and *Faddy* [1974] do apply Markov techniques to the reservoir operation problem. *Chernoff and Petkau* [1978] is a recent representative of this line of inquiry. Although the specific application of their work was to multiprogram computer queues, the structure of the problem is essentially that of the reservoir operations problem. To give a flavor for the ideas involved in the three works just mentioned, we will outline some of the specific developments in the work of *Chernoff and Petkau* [1978], phrasing the variables in reservoir terminology.

Consider a single reservoir with storage $s(t)$ cumulative inflow $w(t)$ and outflow $u(t)$ defined continuously in time (not just on the positive integers, as previously in this survey). It is presumed that the inflow is a Brownian motion with drift θ and variance parameter σ^2. The implications of this assumption which we will need are that for any positive numbers t, t'

$$E[w(t + t') - w(t)] = \theta t' \tag{77}$$

and

$$E[(w(t + t') - w(t))^2] = \sigma^2 t' \tag{78}$$

If the release is maintained at a fixed level u during the time interval (t_1, t_2), then the reservoir storage $s(t)$ changes according to the rule

$$s(t_2) = s(t_1) + [w(t_2) - w(t_1)] - u(t_2 - t_1) \tag{79}$$

There is a 'reflection' modification made if $s(t) = 0$, to prevent the storage from becoming negative. It is presumed that there are only a finite number of allowable release rates $U = \{u^1, u^2, \cdots, u^M\}$ from which the control can be chosen.

We now describe the costs. If, during the time interval (t_1, t_2), the release rate has been held at u^i, then we have the associated cost $c_i(t_2 - t_1)$ and an additional cost reflecting the storage level $s(t)$, given in terms of a specified unit cost function $g(\cdot)$,

$$\int_{t_1}^{t_2} g[s(t)]\, dt$$

Finally, there is a nonnegative cost K_{ij} which is incurred whenever the release rate is switched from u^i to u^j.

With the above continuous time stochastic OCP components specified, we are now in a position to examine the solution proposed by *Chernoff and Petkau* [1978]. The states of the process are $x(t) = (j, s)$, where j indicates that at time t, the release level is u^j, and the reservoir storage volume is s. Let $C(x, t, T)$ denote the expected cost, under an optimal stationary strategy, for controlling the process until time T, beginning at time t and state x. *Chernoff and Petkau* [1978] have proven that for some function $v(x)$ and number γ,

$$C(x, t, T) \simeq v(x) + \gamma(T - t) \tag{80}$$

the approximation becoming exact as $T \to \infty$. The function $v(x)$ is called a 'potential function' and gives a measure of the relative desirability of various starting states. Such a construct shows up in *Howard*'s [1960] work and was exploited in ways similar to that which we will now describe, by *Bather* [1968] and *Faddy* [1974]. The importance of (80) is that if somehow we could compute the function $v(x)$ and the constant γ, then we would have a simple and direct way of describing the optimal stationary strategy, namely, if the state is $x(t) = (j, s)$, continue with control u^j if and only if

$$v(j, s) \le K_{ji} + v(i, s) \qquad 1 \le i \le M \qquad i \ne j \tag{81}$$

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Otherwise switch to $u^{i'}$, where i' is the minimizer of the expression on the right side of (81). The heart of the analysis of Chernoff and Petkau is devoted to computing the potential function. Let us view part of this approach.

Elementary application of the 'principle of optimality' gives that for a small positive number Δ and for every state (i, s) that is a continuation state for control u^i,

$$C[(i, s), t, T] = [c_i + g(s(t))]\Delta + E[C((i, s(t + \Delta)), t + \Delta, T)] + 0(\Delta^2) \quad (82)$$

One now employs the representation (80), taking the first two terms of the Taylor series expansion of $v[(i, s')]$ about $s' = s$, and letting $\Delta s = s(t + \Delta) - s(t)$, to rewrite (82) as

$$v(i, s) + \gamma(T - t) = [c_i + g(s)]\Delta + E[v(i, s) + (v(i, s))_s \Delta s + \tfrac{1}{2}(v(i, s))_{ss}(\Delta s)^2] + \gamma(T - t - \Delta) + 0(\Delta^2) \quad (83)$$

After ignoring second-order terms in Δ and using (77) and (78), we obtain

$$c_i + g(s) - \gamma + \theta[v(i, s)]_s + \tfrac{1}{2}[v(i, s)]_{ss}\sigma^2 = 0 \quad (84)$$

$$l \le i \le m$$

The preceding equation is a linear second-order differential equation system. We will refer the reader to *Chernoff and Petkau* [1978] for a description of ways to obtain the boundary values and for an illustrative computed solution.

One could allow more elaborate Markov models for the inflow process. In particular, the effect of allowing θ and σ^2 to depend on storage would have the effect of making (83) a nonlinear differential equation. This is not a significant drawback if computer solutions are sought.

A disappointing aspect of these developments is that if one wishes to apply this approach to systems of reservoirs, then (84) becomes a partial differential equation with one independent variable for each reservoir. This would seem to limit this technique to systems of three or four reservoirs at most, but even that could be an improvement on the state of the art.

6. Abbreviated List of Symbols

C	reservoir capacity.
DDP	discrete dynamic programming.
DDDP	discrete-differential dynamic programming.
DIFF DP	differential dynamic programming.
D_t	demand at stage t.
$f_t(x, u)$	dynamics function.
$L_t(x, u)$	single-stage loss function.
I	number of discretized controls.
i_t	reservoir inflow at stage t.
IDP	(state) incremental dynamic programming.
J	number of discretized states.
$J(\mathbf{S})$	objective function for stochastic optimal control problem.
$J(\mathbf{u})$	objective function for deterministic optimal control problem.
m	dimension of control vectors.
n	dimension of state vectors.
N	number of decision stages in an optimal control problem.
OCP	optimal control problem.
$p_t(y; x, u)$	transition density function for a stochastic OCP.
s_t	reservoir storage at stage t.
$\mathbf{S} = (S_t)$	stochastic OCP strategy.
$S(x)$	stationary strategy.
t	stage index.
u_t	control at stage t.
$\mathbf{u}$	policy.
$\bar{\mathbf{u}}$	nominal policy.
$\mathbf{u}'$	successor policy.
$V_t(x)$	optimal return function.
x_t	state at stage t.
$\mathbf{x}$	trajectory.
$\bar{\mathbf{x}}$	nominal trajectory.
$x_t(j)$	jth coordinate of state at stage t.

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References

Anis, A., and E. Lloyd, Reservoirs with mixed Markovian-independent inflows, *SIAM J. Appl. Math., 25*(1), 68–76, 1972.

Arrow, K. J., S. Karlin, and H. Scarf, *Studies in the Mathematical Theory of Inventory and Production,* Stanford University Press, Stanford, Calif., 1958.

Arunkumar, S., Characterization of optimal operating policies for finite dams, *J. Math. Anal. Appl., 49,* 267–274, 1975.

Arunkumar, S., Optimal regulation policies for a multipurpose reservoir with seasonal input and return function, *J. Optim. Theory Appl., 21,* 319–328, 1977.

Arunkumar, S., and K. Chon, On optimal regulation for certain multi-reservoir systems, *Oper. Res., 26*(4), 551–562, 1978.

Askew, A. J., Optimum reservoir operating policies and the imposition of a reliability constraint, *Water Resour. Res., 10*(1), 51–56, 1974*a*.

Askew, A. J., Chance-constrained dynamic programming and the optimization of water resources system, *Water Resour. Res., 10*(6), 1099–1106, 1974*b*.

Askew, A. J., Use of risk premium in chance-constrained dynamic programming, *Water Resour. Res., 11*(6), 862–866, 1975.

Askew, A., Comments on 'Reliability-constrained dynamic programming and randomized release rules in reservoir management,' by Lewis A. Rossman, *Water Resour. Res., 14*(1), 159–160, 1978.

Bather, J. A., A diffusion model for the control of a dam, *J. Appl. Probl., 5,* 55–71, 1968.

Becker, L., and W. W.-G. Yeh, Optimal timing, sequencing, and sizing of multiple-reservoir surface water supply facilities, *Water Resour. Res., 10*(1), 57–62, 1974*a*.

Becker, L., and W. W.-G Yeh, Optimization of real time operations of multiple-reservoir system, *Water Resour. Res., 10*(6), 1107–1112, 1974*b*.

Becker, L., W. W.-G. Yeh, D. Fults, and D. Sparks, Operations models for Central Valley project, *J. Water Resour. Plann. Manage. Div. Am. Soc. Civ. Eng., 102*(WRI), 101–115, 1976.

Bellman, R., *Dynamic Programming,* Princeton University Press, Princeton, N. J., 1957.

Bellman, R., and S. Dreyfus, *Applied Dynamic Programming,* Princeton University Press, Princeton, N. J., 1962.

Bertsekas, D., *Dynamic Programming and Stochastic Control,* Academic, New York, 1976.

Bertsekas, D., and S. Shreve, *Stochastic Optimal Control,* Academic, New York, 1978.

Bhat, U., *Elements of Applied Stochastic Processes,* John Wiley, New York, 1972.

19447973, 1982, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/WR018i004p00673 by Ntnu Norwegian University Of S, Wiley Online Library on [04/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Bogle, M., and M. O'Sullivan, Stochastic optimization of a water supply system, *Water Resour. Res.*, *15*(4), 778–786, 1979*a*.
Bogle, M., and M. O'Sullivan, Stochastic optimization of a water supply expansion, *Water Resour. Res.*, *15*(5), 1229–1237, 1979*b*.
Buras, N., Conjunctive operation of dams and aquifers, *Proc. Am. Soc. Civ. Eng.*, *89*(HY6), 111–131, 1963.
Buras, N., *Scientific Allocation of Water Resources*, Elsevier, New York, 1972.
Butcher, W., Stochastic dynamic programming for optimum reservoir operation, *Water Resour. Bull.*, *7*(1), 115–123, 1971.
Butcher, W., Y. Haimes, and W. Hall, Dynamic programming for the optimal sequencing of water supply projects, *Water Resour. Res.*, *5*(4), 1196–1204, 1969.
Chang, S., and W. Yeh, Optimal allocation of artificial aeration along a polluted stream using dynamic programming, *Water Resour. Bull.*, *9*(5), 985–997, 1973.
Chernoff, H., and A. Petkau, Optimal control of a Brownian motion, *SIAM J. Appl. Math.*, *34*(4), 717–731, 1978.
Chow, V. T., D. R. Maidment, and G. W. Tauxe, Computer time and memory requirements for DP and DDDP in water resources systems analysis, *Water Resour. Res.*, *11*(5), 621–628, 1975.
Cohen, J., and M. Rubinovitch, On level crossings and cycles in dam processes, *Math. Oper. Res.*, *2*(4), 297–310, 1977.
Croley, T. E., II, Sequential deterministic optimization in reservoir operation, *J. Hydraul. Div. Am. Soc. Civ. Eng.*, *100*(HY3), 443–459, 1974.
Denny, J., and S. Yakowitz, Admissible run-contingency type tests for independence and Markov dependence, *J. Am. Stat. Assoc.*, *73*, 177–181, 1978.
Denny, J., C. Kisiel, and S. Yakowitz, Procedures for determining the order of Markov dependence in streamflow records, *Water Resour. Res.*, *10*(5), 947–954, 1974.
Derman, C., *Finite State Markovian Decision Processes*, Academic, New York, 1970.
Doran, D. G., An efficient transition definition for discrete state reservoir analysis: The divided interval technique, *Water Resour. Res.*, *11*(6), 867–873, 1975.
Dracup, J., and T. Fogarty, Optimal planning for a thermal discharge treatment system, *Water Resour. Res.*, *10*(1), 67–71, 1974.
Dreyfus, S., and A. Law, The Art and Theory of Dynamic Programming, Academic, New York, 1977.
Dudley, N. J., and O. R. Burt, Stochastic reservoir management and system design for irrigation, *Water Resour. Res.*, *9*(3), 507–522, 1973.
Dudley, N. J., D. T. Howell, and W. F. Musgrave, Optimal intraseasonal irrigation water allocation, *Water Resour. Res.*, *7*(4), 770–788, 1971.
Dudley, N., D. Reklis, and O. R. Burt, Reliability, trade-offs, and water resources development modeling with multiple crops, *Water Resour. Res.*, *12*(6), 1101–1108, 1976.
Dyer, P., and S. McReynolds, *The Computational Theory of Optimal Control*, Academic, New York, 1970.
Dynkin, E., and A. Yushkevich, *Controlled Markov Processes*, Springer-Verlag, New York, 1979.
Eisel, L. M., Chance-constrained reservoir model, *Water Resour. Res.*, *8*(2), 339–347, 1972.
Erlenkotter, D., Sequencing expansion projects, *Oper. Res.*, *21*(2), 542–553, 1973*a*.
Erlenkotter, D., Sequencing of interdependent hydroelectric projects, *Water Resour. Res.*, *9*(1), 21–27, 1973*b*.
Faddy, M. J., Optimal control of finite dams: Continuous output procedure, *Adv. Appl. Probl.*, *6*, 689–710, 1974.
Feller, W., *An Introduction to Probability Theory and Its Applications*, vol. 1, John Wiley, New York, 1968.
Flinn, J., and W. Musgrave, Development and analysis of input-output relations for irrigation water, *Aust. J. Agric. Econ.*, *11*(1), 1–19, 1967.
Fogel, M., L. Duckstein, and C. Kisiel, Optimum control of irrigation with a water application, *J. Hydrol.*, *28*, 343–358, 1976.
Gablinger, M., and D. P. Loucks, Markov models for flow regulation, *J. Hydraul. Div. Am. Soc. Civ. Eng.*, *96*(HY1), 165–181, 1970.
Gagnon, C. R., R. H. Hicks, S. L. S. Jacoby, and J. S. Kowalik, A non-linear programming approach to a very large hydroelectric system optimization, *Math. Program. 6*, 28–41, 1974.
Gal, S., Optimal management of stochastic multireservoir water supply system, *Water Resour. Res.*, *15*(4), 737–749, 1979.
Gani, J., Problems with probability theory of storage systems, *J. R. Stat. Soc. Ser. B*, *19*, 181–206, 1957.
Gessford, J., and S. Karlin, Optimal policy for hydroelectric operations, in *Studies in the Mathematical Theory of Inventory and Production*, edited by K. J. Arrow, S. Karlin, and H. Scarf, pp. 179–200, Stanford University Press, Stanford, Calif., 1958.
Hadley, G., and T. Whitin, *Analysis of Inventory Systems*, Prentice-Hall, Englewood Cliffs, N. J., 1963.
Halkin, H., A maximum principle of Pontryagin-type for systems described by nonlinear difference equations, *SIAM J. Control*, *4*(1), 90–111, 1966.
Hall, W., Aqueduct capacity under an optimum benefit policy, *J. Irrig. Drain. Div. Am. Soc. Civ. Eng.*, *87*(IR3), 1–11, 1961.
Hall, W. A., Optimum design of a multiple-purpose reservoir, *J. Hydraul. Div. Am. Soc. Civ. Eng.*, *90*, 141–149, 1964.
Hall, W. A., and N. Buras, The dynamic programming approach to water resources development, *J. Geophys. Res.*, *66*, 517–521, 1961.
Hall, W., and W. Butcher, Optimal timing of irrigation, *J. Irrig. Drain. Div. Am. Soc. Civ. Eng.*, *94*(IR-2), 267–275, 1968.
Hall, W., and D. Howell, The optimization of single-purpose reservoir design with the application of dynamic programming to synthetic hydrology samples, *J. Hydrol.*, *1*, 355–363, 1963,
Hall, W., and T. Roefs, Hydropower project output optimization, Proceedings, *J. Power Div. Am. Soc. Civ. Eng.*, *PO1*, 67–79, 1966.
Hall, W. A., W. S. Butcher, and A. Esogbue, Optimization of the operation of a multiple-purpose reservoir by dynamic programming, *Water Resour. Res.*, *4*(3), 471–477, 1968.
Hall, W. A., R. Harboe, W. W.-G. Yeh, and A. Askew, Optimum firm power output from a two reservoir system by incremental dynamic programming, *Contrib. 130*, Water Resour. Center, Univ. of Calif. at Los Angeles, Los Angeles, 1969*a*.
Hall, W., G. Tauxe, and W. W.-G. Yeh, An alternate procedure for the optimization of operations for planning with multiple-river, multiple-purpose systems, *Water Resour. Res.*, *5*(6), 1367–1372, 1969*b*.
Haussmann, U. G., Some examples of optimal stochastic controls or: The stochastic maximum principle at work, *SIAM Rev.*, *23*(3), 292–307, 1981.
Heidari, M., V. T. Chow, P. V. Kokotovic, and D. D. Meredith, Discrete differential dynamic programming approach to water resources systems optimization, *Water Resour. Res.*, *7*(2), 273–282, 1971.
Hinderer, K., *Foundations of Non-Stationary Dynamic Programming With Discrete Time Parameter*, Springer-Verlag, New York, 1970.
Howard, R. A., *Dynamic Programming and Markov Processes*, MIT Press, Cambridge, Mass., 1960.
Howard, R., Comments on the origin and application of Markov decision processes, in *Dynamic Programming and Its Applications*, edited by M. Puterman, Academic, New York, 1978.
Jacobson, D., and D. Mayne, *Differential Dynamic Programming*, Academic, New York, 1970.
Jacoby, S., and J. Kowalik, *Mathematical Modelling with Computers*, Prentice-Hall, Englewood Cliffs, N. J. 1980.
Joeres, E. F., J. C. Leibman, and C. S. Revelle, Operating rules of joint operations of water sources, *Water Resour. Res.*, *7*(2), 225–235, 1971.
Kaplan, M., and Y. Haimes, Dynamic programming for optimal capacity expansion of wastewater treatment plants, *Water Resour. Bull.*, *11*(2), 278–293, 1975.
Kendall, D., Some problems in the theory of dams, *J. R. Stat. Soc. Ser. B*, *19*, 207–212, 1957.
Kirk, D., *Optimal Control Theory*, Prentice-Hall, Englewood Cliffs, N. J., 1970.
Klemeš, V., Discrete representation of storage for stochastic reservoir optimization, *Water Resour. Res.*, *13*(1), 149–158, 1977.
Korsak, A., and R. Larson, A dynamic programming successive approximations technique with convergence proofs, *Automatica*, *6*, 245–252, 1970.
Labadie, J., N. Grigg, and P. Trotta, Minimization of combined sewer overflows by large-scale mathematical programming, *Comput. Oper. Res.*, *1*, 421–435, 1974.
Labadie, J., N. Grigg, and B. Bradford, Automatic control of large-scale combined sewer systems. *J. Environ. Eng. Div. Am. Soc. Civ. Eng.*, *101*, 27–39, 1975.

19447973, 1982, 4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/WR018i004p00673 by Nmu Norwegian University Of S, Wiley Online Library on [04/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Larson, R., *State Increment Dynamic Programming*, Elsevier, New York, 1968.
Larson, R., and J. Casti, *Principles of Dynamic Programming*, vol 1, Dekker, New York, 1978.
Lasdon, L., and A. Waren, Survey of nonlinear programming applications, *Oper. Res., 28*, 1029–1073, 1980.
Leibman, J., and W. Lynn, The optimal allocation of stream dissolved oxygen, *Water Resour. Res., 2*(3), 581–591, 1966.
Lindgren, B., *Elements of Decision Theory*, Macmillan, New York, 1971.
Little, J. D. C., The use of storage water in a hydroelectric system, *Oper. Res., 3*, 187–197, 1955.
Lloyd, E., Stochastic reservoir theory, *Adv. Hydrosci., 4*, 281–339, 1967.
Lloyd, E., Some aspects of stochastic reservoir theory—A further comment, *J. Hydrol., 36*, 187–191, 1978.
Masse, P. B. D., *Les Reserves et la Régulation de l'Avenir dans la Vie Economique*, Hermann and Cie, Paris, 1946.
Matanga, G., and M. Mariño, Irrigation planning, 2, Water allocation for teaching and irrigation purposes, *Water Resour. Res., 15*(3), 679–683, 1979.
Mawer, P., and D. Thorn, Improved dynamic programming procedures and their practical application, *Water Resour. Res., 10*(2), 183–190, 1974.
Mays, L., and H. Wenzel, Optimal design of multilevel branching sewer systems, *Water Resour. Res., 12*(5), 913–917, 1976.
Meier, W. L., Jr., and C. S. Beightler, An optimization method for branching multistage water resource systems, *Water Resour. Res., 3*(3), 645–652, 1967.
Moore, N., and W. Yeh, Economic model for reservoir planning, *J. Water Resour. Plann. Manage. Div. Am. Soc. Civ. Eng., 106*(WR2), 383–400, 1980.
Moran, P., A probability theory of dams and storage systems, *Aust. J. Appl. Sci., 5*, 116–124, 1954.
Moran, P., *The Theory of Storage*, John Wiley, New York, 1959.
Morin, T. L., Pathology of a dynamic programming sequencing algorithm, *Water Resour. Res., 9*(5), 1178–1185, 1973.
Morin, T., Optimality of a heuristic sequencing technique, *ASCE J. Hydraul. Div. Am. Soc. Civ. Eng., 100*(HY8), 1195–1202, 1974.
Morin, T., and A. Esogbue, Some efficient dynamic programming algorithms for the optimal sequencing and scheduling of water supply projects, *Water Resour. Res., 7*(3), 479–484, 1971.
Murray, D., and S. Yakowitz, Constrained differential dynamic programming and its application to multireservoir control, *Water Resour. Res., 15*(4), 1017–1027, 1979.
Murray, D., and S. Yakowitz, The application of optimal control theory to nonlinear programming, *Math. Program., 21*, 331–347, 1981.
Nemhauser, G. L., *Dynamic Programming*, John Wiley, New York, 1966.
Nopmongcol, P., and A. Askew, Multilevel incremental dynamic programming, *Water Resour. Res., 12*(6), 1291–1297, 1976.
O'Laoghaire, D., and D. Himmelblau, *Optimal Expansion of a Water Resources System*, Academic, New York, 1974.
Puterman, M. (Ed.), *Dynamic Programming and Its Applications*, Academic, New York, 1978.
Revelle, C., E. Joeres, and W. Kirby, The linear decision rule in reservoir management and design, 1, Development of the stochastic model, *Water Resour. Res., 5*(4), 767–777, 1969.
Rhenals, A., and R. L. Bras, The irrigation scheduling problem and evapotranspiration uncertainty, manuscript, Parsons Lab., Mass. Inst. of Technol., Cambridge, 1981.
Riordan, C., A. general multistage marginal cost dynamic programming model for the optimization of a class of investment-pricing decisions, *Water Resour. Res., 7*(2), 245–253, 1971.
Rockafellar, R. T., and R. Wets, The optimal recourse problem in discrete time: L^1-multipliers for inequality constraint, *SIAM J. Control Optim., 16*, 16–36, 1978.
Roefs, T. G., and L. D. Bodin, Multireservoir operation studies, *Water Resour. Res., 6*(2), 410–420, 1970.
Rosenthal, R., The status of optimization models for the operation of multireservoir systems with stochastic inflows and nonseparable benefits, *Rep. 75*, Tenn. Water Resour. Res. Center, Knoxville, May 1980.
Rossman, L., Reliability-constrained dynamic programming and randomized release rules in reservoir management, *Water Resour. Res., 13*(2), 247–255, 1977.
Russell, C. B., An optimal policy for operating a multipurpose reservoir, *Oper. Res., 20*(6), 1181–1189, 1972.
Sathaye, J., and W. Hall, Aqueduct optimization with intermediate storage, *J. Irrig. Drain. Div. Am. Soc. Civ. Eng., 102*(IR2), 249–264, 1976*a*.
Sathaye, J., and W. A. Hall, Optimization of design capacity of an aqueduct, *J. Irrig. Drain. Div. Am. Soc. Civ. Eng., 102*(IR3), 295–305, 1976*b*.
Schweig, Z., and A. Cole, Optimal control of linked reservoirs, *Water Resour. Res., 4*(3), 479–498, 1968.
Sniedovich, M., Reliability-constrained reservoir control problems, 1, Methodological issues, *Water Resour. Res., 15*(6), 1574–1582, 1979.
Sniedovich, M., A variance-constrained reservoir control problem, *Water Resour. Res., 16*(2), 271–274, 1980.
Sniedovich, M., Some comments on preference order dynamic programming models, *Math. Anal. Appl., 79*(2), 489–501, 1981.
Sniedovich, M., and D. R. Davis, Comments on 'Chance-constrained dynamic programming and optimization of water resource systems' by Authur J. Askew, *Water Resour. Res., 11*(6), 1037–1038, 1975.
Sobel, M. J., Reservoir management models, *Water Resour. Res., 11*(6), 767–777, 1975.
Srinivasan, S., Analytic solution of a finite dam governed by a general input, *J. Appl. Probl., 11*, 134–144, 1974.
Su, S. Y., and R. A. Deininger, Modeling the regulation of Lake Superior under uncertainty of future water supplies, *Water Resour. Res., 10*, 11–25, 1974.
Tamura, H., Multistage decomposition algorithms for optimizing discrete dynamic systems with applications, *Handbook of Large Scale Systems Engineering Applications*, edited by M. Singh and A. Titli, North-Holland, Amsterdam, 1979.
Tauxe, G., R. Inman, and D. Mades, Multiobjective dynamic programming with applications to a reservoir, *Water Resour. Res., 15*(6), 1403–1408, 1979*a*.
Tauxe, G., R. Inman, and D. Mades, Multiobjective dynamic programming, A classic problem redressed, *Water Resour. Res., 15*(6), 1398–1402, 1979*b*.
Trott, W., and W. Yeh, Optimization of multiple reservoir system, *J. Hydraul. Div. Am. Soc. Civ. Eng., 99*(HY10), 1865–1884, 1973.
White, D. J., Dynamic programming and probabilistic constraints, *Oper. Res., 22*(3), 654–664, 1974.
Yakowitz, S., *Mathematics of Adaptive Control Processes*, Elsevier, New York, 1969.
Yakowitz, S., A stochastic model for daily river flows in an arid region, *Water Resour. Res., 9*(5), 1271–1285, 1973.
Yakowitz, S., Small sample hypothesis, tests of Markov order, with application to simulated hydrologic chains, *J. Am. Stat. Assoc., 71*(353), 132–136, 1976.
Yakowitz, S., A nonparametric Markov model for daily river flow, *Water Resour. Res., 15*(5), 1035–1043, 1979.
Yakowitz, S., Convergence bounds for the dynamic programming successive approximations method, submitted to *Automatica*, 1981.
Yakowitz, S., Constrained differential dynamic programming, submitted to *Math. Anal. Appl.*, 1982.
Yakowitz, S., and B. Rutherford, Contributions to discrete-time differential dynamic programming, submitted to *Optimal Contr.*, 1981.
Yeh, W., and L. Becker, Real-time hourly reservoir operation, *J. Water Resour. Plann. Manage. Div. Am. Soc. Civ. Eng., 105*(WR2), 187–303, 1979.
Young, G. K., Finding reservoir operating rules, *J. Hydraul. Div. Am. Soc. Civ. Eng., 93*(HY6), 297–321, 1967.

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